
Typical: Typed Decisions Read Directly from Pretrained Language Models

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Abstract

Software that asks a language model for a decision needs a probability for each option it supplies, and a way to say that none of them fit. Typical reads that distribution directly from a pretrained language model’s hidden states, without generating text. It encodes the input document once, reads each question with its candidate options in a short suffix, and scores every candidate at 71% of the network’s depth, returning a choice, an ordinal score or a yes/no probability, plus a separate probability of abstaining. The hard part of building such a model is making the decision depend on the right inputs. A model can score well while ignoring the question, the evidence or the rubric, or while shifting with label order and with how many options it is offered, and accuracy shows none of it. We built Typical around tests that do. A shuffled-question control, Δ_q , shows that question-dependent knowledge appears only when the candidates are read inside the forward pass, in every configuration we tested. Training on counterfactual rubric groups makes the model use the rubric, lifting held-out rule families by 35–40 points over a model never trained on them, and when training truncates long documents, placing the evidence first adds 11.2 points. A Bernoulli head makes yes/no answers invariant to label order, and removing the “none of the above” option from the candidates, leaving abstention to a separate gate, cuts MMLU-Pro false abstention from .692 to .003. A decision takes 45 ms at 1.7B and 60 ms at 14B, and the 14B model reaches .931 on the JevBench standard tier. We release 1.7B and 4B models and an inference package.

1 Introduction

Many language-model calls in production software ask for a decision: which of forty queues a ticket belongs in, whether a transaction is fraud, how severe an alert is, which tool an agent should call next. The caller needs a probability for each option it defines at request time, and it needs to know when none of them fit. Today that distribution is recovered from text, by generating an answer symbol or a JSON object, constraining decoding, or comparing candidate log-probabilities. All of these pay for decoding and parsing, and all inherit the generative interface’s sensitivity to how the answer is written (Zheng et al., 2023; Pezeshkpour and Hruschka, 2024; Wang et al., 2024a).

When a causal language model reads a question followed by a list of options, the winning option is already decodable from its hidden states before the model writes an answer symbol (Wong et al., 2026). Typical turns that internal decision into the interface. A call maps

$$(\text{state}, \text{typed question}, \text{candidates}) \longmapsto (P(a_1), \dots, P(a_K), P(\emptyset))$$

with nothing generated or parsed. The state (a ticket, a policy, an agent trace) is encoded once into a key-value cache. Each question is a short causal suffix holding the question and its candidates, and a head at 71% of the backbone’s depth scores the decision state against each candidate’s contextual

Table 1: What a decision should and should not depend on. Each row pairs a test with the failure it exposed in a naive design and the result with Typical’s design. Section numbers point to the evidence.

Input or factor	Test	Naive design	Typical
<i>The decision should depend on</i>			
the question	shuffled question, Δ_q	state computed before the candidates: $ \Delta_q < .03$	candidates read in the forward pass: .09–.12 (§5.1)
the evidence	same head, last layer vs. 71% depth	last layer: MNLI .752, ANLI .431	71% tap: .865, .519 (§5.2)
evidence in long documents	training $\times$ evaluation render	facts truncated in training: .830	evidence first: .942 (§5.2)
the rubric	held-out rubric-flip pairs	never trained on rubric groups: .477	counterfactual rubric groups: .710 (§5.3)
<i>The decision should not depend on</i>			
label order	reverse the yes/no labels	two-way Choice: $\Delta P(\text{yes})$ up to .55	Bernoulli Noul: 0 (§6.1)
options offered	vary K with gold absent	rendered “none of the above”: MMLU-Pro false abstention .692	gate only: .003 (§6.2)
formatting	swap the layout	frozen 9B: 12.5 points from layout alone	trained 1.7B: 14.5–19.0 on mismatch; 14B: 6.8–7.6 (§6.3)

hidden state (Figure 1). The output type is a contract on the distribution: categorical *Choice*, ordinal *Score*, a Bernoulli yes/no decision we call *Noul*, and an abstention probability $P(\emptyset)$ computed by a gate outside the candidate softmax.

Reading the decision out turned out to be the easy part. The hard part is making it depend on the right inputs. A decision model can reach high accuracy while answering from which options usually go together instead of from the question, from the hypothesis instead of the evidence, or from the state instead of the rubric. It can also shift when the labels are reordered, when more options are offered, or when the same content is laid out differently. Accuracy shows none of this, because a model that leans on the wrong input is often right anyway.

We therefore built Typical around tests of what its decisions depend on. For each input that should decide the answer we use a control that removes it, and for each factor that should not we use a perturbation that varies it (Section 4). Table 1 lists what each test found in a naive design and what Typical’s design does instead. Six design choices trace to its rows: reading candidates inside the forward pass, the 71% tap, counterfactual rubric training, evidence-first long documents, the Bernoulli head and the abstention gate. Section 7 covers probability quality and cost.

The resulting models are fast. A decision takes 45 ms at 1.7B and 60 ms at 14B in a research harness, because the cached state and a short suffix dominate its cost, and the 14B model reaches .931 on the JevBench standard tier (Section 7).

Contributions.

1. A direct decision interface for pretrained causal language models, with typed outputs and a factored abstention gate, that returns distributions over runtime candidates without generating text (Section 3).
2. A protocol for testing what a decision model depends on, built on Δ_q and render-crossed evaluation, which applies to any architecture that scores candidates (Section 4).
3. Design and training choices each tied to a failure that protocol exposed (Sections 5 and 6).
4. Open models, typical-small (1.7B) and typical-medium (4B), with an inference package (`pip install typical-ai`).

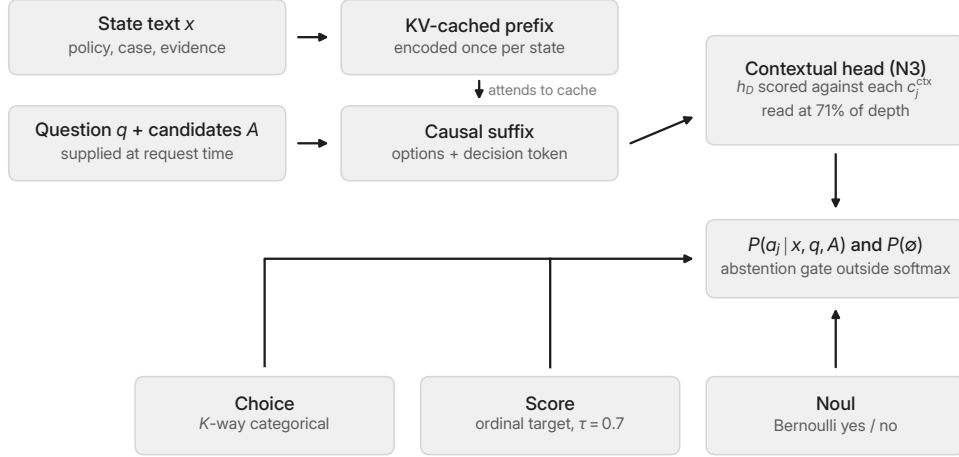


Figure 1: The decision pass. The state is encoded once into a key-value cache. Each question is a short causal suffix holding the question and its runtime candidates; the head reads the terminal decision state h_D against each candidate’s pooled contextual state c_j^{ctx} at 71% of backbone depth. One readout serves three typed contracts (Choice, Score, Noul), and abstention is a separate gate outside the candidate softmax.

2 Related work

Decisions in hidden states. Probes recover linearly decodable structure from hidden states (Alain and Bengio, 2016), and that structure changes with depth (Tenney et al., 2019; Belrose et al., 2023). Circuit analysis localizes where a multiple-choice model binds its chosen option to a symbol (Lieberum et al., 2023), and Wong et al. (2026) decode the winning option in content space before that binding happens, a result Bhatt et al. (2026) trace across languages. Typical deploys the pre-symbol decision as the interface and reads it at mid-depth, where upper layers are partly redundant (Gromov et al., 2024; Men et al., 2025).

Answer symbols and runtime labels. Letter answers carry order and symbol biases (Zheng et al., 2023; Pezeshkpour and Hruschka, 2024), first-token probabilities disagree with the text a model generates (Wang et al., 2024a), and format constraints can cost accuracy (Tam et al., 2024). Δ_q (Definition 4.1) extends the choices-only control of Balepur et al. (2024) to a shuffled real question. Runtime labels have been scored through entailment (Yin et al., 2019), joint schema encoders (Zarariana et al., 2025), cross-encoders (Nogueira and Cho, 2019), dual encoders (Reimers and Gurevych, 2019) and late interaction (Humeau et al., 2019; Khattab and Zaharia, 2020), and decoder classification heads fix labels as indices (Li et al., 2023). Typical renders all candidates in one causal suffix, so each candidate’s state carries its slot and its set, and Δ_q measures what candidate-independent scoring gives up (Section 5.1).

Computation per decision. Prefix caching and sharing are established serving techniques (Kwon et al., 2023; Zheng et al., 2024; Gim et al., 2023), and speculative decoding shortens the decode loop (Leviathan et al., 2022); Typical has no decode loop. Chain-of-thought supplies the serial computation that a fixed-depth pass lacks (Wei et al., 2022; Merrill and Sabharwal, 2023a; Feng et al., 2023; Merrill and Sabharwal, 2023b), as do pause tokens (Goyal et al., 2023) and latent reasoning (Hao et al., 2024); Section 8 connects this to the decisions Typical finds hardest.

Output contracts and training. Score’s soft ordinal target follows ordinal regression (Frank and Hall, 2001; Cao et al., 2020). The abstention gate descends from the reject option (Chow, 1970; Geifman and El-Yaniv, 2017) and confidence-based out-of-scope detection (Hendrycks and Gimpel, 2016; Larson et al., 2019). We evaluate probabilities with proper scoring rules (Gneiting and

Raftery, 2007) alongside temperature-scaled calibration (Guo et al., 2017), and train with LoRA (Hu et al., 2021) and distillation (Hinton et al., 2015). Compositional generalization benchmarks (Lake and Baroni, 2017; Keysers et al., 2019) motivate the held-out composition level of Decision-Mix (Section 3.3), and work on evidence position in long contexts (Liu et al., 2024) and silent data-pipeline defects (Sambasivan et al., 2021) motivates the render-crossed design (Section 5.2). Among JevBench’s open entries, only GLiNER2 (Zaratiana et al., 2025) and a ModernBERT-based model (Warner et al., 2025) have method papers.

3 Typical

Typical turns a pretrained causal language model into a decision function whose output is read from hidden states in one pass over a short suffix.

3.1 The decision pass

Definition 3.1 (Typed decision). A decision is a tuple $D = (x, q, A, t)$: a state x (the document, transcript or trace the decision is about), a question q (the instruction or rubric), a candidate set $A = \{a_1, \dots, a_K\}$ of label strings supplied at request time, and a type $t \in \{\text{CHOICE}, \text{SCORE}, \text{NOUL}\}$. The model returns

$$P(a_j \mid x, q, A), \quad j = 1, \dots, K, \quad P(\emptyset) = P(\emptyset \mid x, q, A), \quad P(\emptyset) + \sum_j P(a_j \mid x, q, A) = 1,$$

and emits no token.

The abstention event $\emptyset$ is available under every type and is never one of the candidates. Because A is text supplied per call, the model has no output index for any label and must read the candidates to score them.

Cached state and causal suffix. Typical encodes the state once as an ordinary causal prefix and retains its key-value cache. Each question is a short causal suffix: the question text, the candidates rendered as lettered lines, an answer cue, and a terminal decision token. A decision is therefore a forward pass on $\text{concat}(x, q, A)$ with the state half read from cache (Appendix A.20).

Contextual-candidate readout. At the tap layer the suffix pass yields the terminal decision state h_D , the hidden state at the decision token, and for each candidate a contextual state c_j^{ctx} , the mean hidden state over the tokens of candidate j ’s rendered span. Candidate j receives the score

$$u = W_h h_D, \quad v_j = W_c c_j^{\text{ctx}}, \quad s_j = \frac{u^\top v_j}{\sqrt{d_v}} + w^\top (u \odot v_j) + g(u \odot v_j), \quad (1)$$

where W_h and W_c are layer-normalized linear projections to width d_v , w is a vector and g a two-layer MLP. The option letters stay in the rendered text and are never read out. Because the candidates are tokens in the same causal computation as the state and the question, c_j^{ctx} carries the candidate’s meaning together with its slot and its set, and h_D has attended to all of them. Section 5.1 compares *N1*, the letter-logit readout (the next-token logits of the option letters at h_D , with a softmax null), *N2*, the candidate-blind semantic readout (Equation (1) on a frozen text embedding of each candidate), and *N3*, the contextual-candidate readout Typical deploys (Equation (1) on c_j^{ctx}); Appendix A.19 lists what each pair shares.

Mid-depth tap. We truncate the backbone at about 71% of its depth: the readout taps layer 20 of 28 at 1.7B, 26 of 36 at 4B and 8B, and 28 of 40 at 14B, and the layers above the tap are never computed. LoRA (Hu et al., 2021) at rank 16 adapts the top eight kept layers, and everything below is frozen. The tap carries evidence and question-conditioned knowledge in one model, where the final layer does not (Section 5.2), and every released and ladder model uses it.

Cost. M decisions against one state cost $O(|x|) + M \cdot O(|q| + |A|)$ token positions, where $|A|$ counts the rendered candidate tokens, and every position runs through only the kept 71% of the layers. The state term is paid once; the per-decision term grows with the question and the candidate text and is independent of the other questions. At large K the rendered candidates dominate it, and in throughput mode a candidate-blind scorer is cheaper per query by a margin that grows with K (Section 7.2).

3.2 Typed outputs and abstention

The type is a contract on the returned distribution, enforced by the head and the training target. All three types share the backbone and the tap. *Choice* is categorical over A : the K -way readout of Equation (1), trained with cross-entropy against the gold distribution over $A \cup \{\emptyset\}$ through the gate below. *Score* is categorical over ordered levels, where near misses cost less than distant ones: it uses the Choice readout with an ordinal-smoothed target $t_j \propto \exp(-|j - y|/\tau)$ for gold level y , normalized over the valid levels, with $\tau = 0.7$; $\tau \rightarrow 0$ recovers the one-hot Choice target, and the change adds no parameters. *Noul* is a yes/no decision with $A = \{\text{yes}, \text{no}\}$: a dedicated Bernoulli head, $P(\text{yes}) = \sigma(w_n^\top h_D)$, on a query-only suffix that renders no candidate text, with the complement assigned to “no” and its own label-symmetric gate, so the returned yes, no and $\emptyset$ probabilities are exactly invariant to label order. Routing is per row, because a batch mixes types; rows that are not Noul or Score are scored by the Choice path, bit-identical to a model with the typed heads disabled (Appendices A.7 and A.19).

Factored abstention gate. Abstention is a separate gate outside the candidate softmax. The gate reads permutation-invariant statistics of the scores s_j , the projected decision state and the moments of the projected candidate states v_j (Appendix A.19), and computes $r = \sigma(g_\emptyset(z))$. The returned distribution is

$$P(a_j \mid x, q, A) = (1 - r) \frac{\exp(s_j/T)}{\sum_{k=1}^K \exp(s_k/T)}, \quad P(\emptyset) = r, \quad (2)$$

trained by one cross-entropy over $K + 1$ outcomes. Two properties follow exactly. (i) *Pairwise non-null odds are invariant to abstention*: for fixed scores $P(a_j)/P(a_k) = \exp((s_j - s_k)/T)$, independent of r ; the contextual scores themselves remain set-dependent. (ii) *The gate is temperature-invariant*: T enters only the candidate softmax and r reads untempered statistics, so recalibrating T never moves $P(\emptyset)$. The gate adds no order dependence, while r depends on what was offered by design. We never render the null as a candidate line; Section 6.2 removes that line from a model that already has the gate and measures the difference.

3.3 Training

Every row is a typed decision rendered as above and trained with cross-entropy against a hard or soft gold distribution over $A \cup \{\emptyset\}$, plus distillation at 14B, using AdamW in bf16 at batch size 64 for 8,000–12,000 steps (Appendix A.9). Table 27 lists the seven corpora, and Section 6 measures the six recipe choices below.

Family-balanced mixture. We draw batches by decision family, so no large corpus dominates by row count: E is evidence and wide-vocabulary classification, K closed-book multiple-choice knowledge, W rubric-conditioned workflow decisions, and U soft-target uncertainty (sources in Appendix A.19). The released recipe samples $E:0.40$, $K:0.15$, $W:0.35$, $U:0.10$.

Counterfactual rubric groups. DecisionMix v2 (data_wh) is a programmatic rule-engine corpus in which every row belongs to a *counterfactual rubric group*: the same state and candidate set under at least two rubrics with distinct gold answers. A predictor that ignores the rubric assigns one distribution to every member of a group, so on a two-member group with golds $y_1 \neq y_2$ it incurs an average cross-entropy of at least $\log 2$ nats, and the loss falls below that floor only by reading the rubric. Held-out compositions, domains, styles and grammars are evaluation-only (Appendix A.10).

Soft-target uncertainty. Decisions with a single correct answer cannot teach a model how much probability to spread. The uncertainty corpus data_u draws on UNLI (Chen et al., 2020), the ambiguous rows of AmbiEnt, and four synthetic generators with exact closed-form posteriors (Table 15); ChaosNLI (Nie et al., 2020b) is evaluation-only.

Null augmentation. A copy of 20% of W rows drops the gold candidate and any rendered catch-all option and takes $\emptyset$ as its target. Under 1% of W rows carry a null target against over 20% of E rows, so a model trained on both learns two operating points that no single post-hoc temperature and null offset can serve.

Evidence-first long states. Training right-truncates states at the window, so rendering the case facts after the policy body in the long-state corpus `data_wf_long` lost the facts from most rows (Appendix A.21). The fixed recipe renders facts first and drops any row longer than the window. The v2 releases train at a 2,048-token window and the 14B model at 3,072.

Selection on held-out likelihood. Selection on in-distribution loss favors the hard-label majority of the mixture over the soft and ordinal minority, so the v2 releases and the 14B model select checkpoints on held-out negative log-likelihood over the uncertainty and curriculum validation sets; the v1 releases do not. The 14B model also distills from its own frozen backbone (Appendix A.9). Selection enters the 14B recipe together with evidence-first long states, the 3,072-token window, DecisionMix v2, the uncertainty corpus and the typed heads, and is never isolated.

4 Testing what a decision depends on

A decision model can score well while reading the wrong parts of its input, so we tested each part of Section 3 with one protocol. For each input that should decide the answer (the question, the evidence, the rubric) a control removes it and we measure how much accuracy falls. For each factor that should leave the answer alone (label order, the number of options offered, formatting) a perturbation varies it and we measure how far the output moves. Accuracy on the unperturbed input measures neither.

Inputs that should matter. For the question, the control replaces the real question and holds the state and candidates fixed.

Definition 4.1 (Question-dependence gap). $\Delta_q = \text{Acc}(f(x, q, A)) - \text{Acc}(f(x, \tilde{q}, A))$, where $\tilde{q}$ replaces the real question with the state and candidate set held fixed: an empty placeholder in the *choices-only* variant, a real unrelated question from the same corpus in the *shuffled-question* variant Δ_q^{sh} , primary throughout.

Δ_q extends the choices-only control of Balepur et al. (2024) to a shuffled real question, which is stricter because a placeholder can leak candidate priors through the rendering. The reference is the listwise teacher at $\Delta_q^{\text{sh}} = .102$ on the MMLU-Pro probe, and $|\Delta_q^{\text{sh}}| < .02$ counts as zero (Appendix A.1). For the evidence, SNLI accuracy is set against the hypothesis-only baseline, and long documents are rendered with the facts before or after the policy text, with training render crossed against evaluation render. For the rubric, *rubric-flip* pairs share a state and candidate set under two rubrics with different gold, and a shuffled rubric replaces the real one.

Factors that should not matter. For label order we reverse the labels of a yes/no decision, and on MMLU-Pro we reorder the candidates (*reorder Δp*) or add three unrelated options (*pairwise $\Delta \log\text{-odds}$* ; Appendix A.1). For the number of options we remove the gold label, sweep K and record how often the model abstains. For formatting we change only the layout of the state and options, with weights and items fixed, and read the off-diagonal cells of the render crossing.

Evaluation sets and JevBench. Training corpora are in Table 27; the evaluation sets, held-out splits, metrics, bootstraps and latency protocol are in Appendices A.1 and A.14. JevBench (fstandhartinger/jevbench v1.2.1) is a third-party benchmark of yes/no, choice and ordinal decisions in standard, easy and hard tiers, and every JevBench number here is on its 231 public ids and unranked.

5 What the decision should depend on

A decision should follow the question it is asked, the evidence it is given and the rubric it is told to apply. Each subsection takes one of these inputs, shows how a natural design lets the decision drift away from it, and gives the control that exposed the drift and the change that repaired it. Unless stated otherwise, models here are 1.7B, trained for 12k steps on the evidence and knowledge mixture, and Δ_q^{sh} is measured on the MMLU-Pro probe against the teacher’s .102.

Table 2: Three readouts of one decision pass, full-depth 1.7B backbone, identical data and steps; N1 and N3 at two seeds (s0/s1), N2 at one. Order measures are zero for a set-invariant scorer.

	N1 s0	N1 s1	N2	N3 s0	N3 s1	Teacher
Δ_q^{sh} (primary)	.114	.104	.058	.117	.094	.102
MMLU-Pro among- K	.325	.318	.198	.314	.301	.310
Banking77, $K=77$	.531	—	.012	.497	—	—
SNLI	.838	.831	.854	.863	.851	—
MNLI	.684	.664	.742	.752	.719	—
ANLI	.377	.393	.410	.431	.437	—
BoolQ	.769	.764	.775	.737	.746	—
Reorder Δp	.165	.165	.059	.101	.109	.097
Pairwise $\Delta \log\text{-odds}$	.463	.443	.379	.126	.094	.179

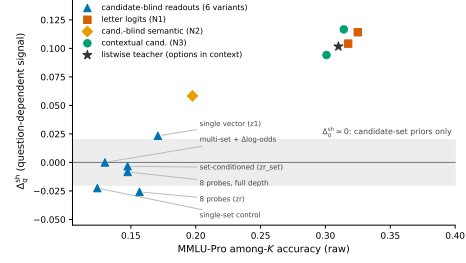


Figure 2: MMLU-Pro among- K accuracy against Δ_q^{sh} ($n=1,200$; shaded: the ± 0.02 band). The six candidate-blind variants cluster at zero; N1, N3 and the teacher carry .09–.12.

5.1 The question

The decision should change when the question changes. The design that fails this test is the one that is cheaper to serve in throughput mode (Section 7.2): a candidate-blind variant computes a decision state $Z(x, q)$ before the candidates are seen, scores each candidate against it with a learned head, and is distilled from the listwise teacher. Accuracy hides the gap because candidate-blind states learn priors and calibration. Their accuracy of .12–.17 sits above the .10 chance rate and comes from which option strings tend to be correct together (Appendix A.2).

Δ_q^{sh} exposes the failure. In every configuration we tested, question-conditioned capability appeared only when the candidates were processed inside the forward pass. Six candidate-blind variants, spanning three head factorizations, two distillation objectives and two tap depths, all have Δ_q^{sh} within .03 of zero, while N1, N3 and the teacher sit at .09–.12 and even N2 keeps .058 (Figure 2; per-variant numbers in Table 6).

No lever we varied moves Δ_q^{sh} , including cross-attention over all candidate tokens, multi-set log-odds supervision and full-depth reading, though we cannot exclude optimization (Appendix A.2). The question-dependent component lives in candidate-contextual computation, so Typical renders the candidates inside the backbone’s forward pass. Any architecture that scores candidates against a precomputed representation should report Δ_q alongside accuracy.

Rendering the candidates leaves open what the decision state is scored against. Table 2 compares the readouts of Section 3.1 on one pass: letter logits (N1), h_D against an independent embedding of each option’s text (N2), and h_D against each option’s contextual state c_j^{ctx} (N3). All arms render the candidates in the suffix, train on identical data with per-step option shuffling, and read all 28 layers, so N2 and N3 differ only in what h_D is scored against. The candidate’s contextual state carries the decision. N2 loses 10–12 points of among- K accuracy against N3 and 13 against N1, half of the question-conditioned signal (Δ_q^{sh} .058 vs. .106 for N3 across seeds), and collapses on Banking77 at $K=77$ (.012). The pattern fits a decision state that encodes which rendered slot holds the answer: the letter embedding and the contextual state both carry slot identity, and an independently embedded label carries only its meaning. N3 matches N1 on question dependence within seed noise: across seeds N1 averages $\Delta_q^{\text{sh}} = .109$ and N3 .106, within the .01–.02 seed spread of both, and among- K accuracy is 1–2 points lower for N3. On evidence tasks N3 leads in both seeds by 2 points on SNLI, 5.5–7 on MNLI and 4–5 on ANLI, and trails by 2–3 on BoolQ. N3 is the readout Typical deploys.

5.2 The evidence

The decision should change when the evidence changes. Two designs failed this test in different places, one in the architecture and one in the training data. The first full-scale model in this project read the last layer and scored 58.6 on SNLI, the score of a baseline that never sees the premise (Appendix A.2). Comparing against that hypothesis-only baseline is what exposed it. Reading at 71% of depth gives one model that carries both evidence reasoning and question-conditioned knowledge. With head, data and steps fixed, reading N3 at layer 20 instead of 28 raises SNLI/MNLI/ANLI/BoolQ from .863/.752/.431/.737 to .906/.865/.519/.834 while Δ_q^{sh} stays at .118

Table 3: Training render $\times$ evaluation render on 605 held-out long states, no truncation at evaluation. Diagonal (matched) cells score each model in its training render; off-diagonal cells hold the evaluation render fixed across the two models. Floors and sizes in Appendix A.1. 14B arms: `t11b_nokd` (facts-first) and `ladder_14b` (facts-last).

Trained on	Scored on	1.7B, single-variable (seed 0)			14B corroboration		
		overall	permit	select	overall	permit	select
facts-first	facts-first (<i>matched</i>)	.942	.933	.953	.997	.997	.996
facts-first	facts-last	.797	.788	.807	.921	.933	.905
facts-last	facts-first	.640	.615	.669	.851	.839	.865
facts-last	facts-last (<i>matched</i>)	.830	.842	.815	.919	.930	.905

against .117 (CLINC-150 and validation NLL in Appendix A.2). Evidence accuracy lands within 1–4 points of a candidate-blind energy-scoring baseline on every evidence set and 8 points above it on CLINC-150, so one model serves the evidence and knowledge regimes that had needed separate models. For a candidate-blind state the same change moves nothing on Δ_q^{sh} , and its evidence accuracy regresses at full depth (Table 7).

The second failure sat in the long-document training data. Training right-truncates states at the window, and the long-state corpus rendered the case facts after the policy text, so at 1,024 tokens the facts were lost in 98.8% of long-policy rows (Section 3.3). The test that exposed the defect is a render-crossed 2×2 . Two 1.7B arms with byte-identical flags train on the same 23,318 rows, rendered facts-first or facts-last, with truncation active as at release, and are scored under both renders (Table 3) on 605 held-out long states, never truncated at evaluation, that differ only in where the evidence sits (Appendix A.15). The comparison is a *matched-configuration gap*: each model is scored in the render it was trained on, so the evaluation render changes along with training.

Training with the evidence placed first adds 11.2 points on held-out long states when training right-truncates states at 1,024 tokens, and the gap replicates at a second seed. At 1.7B the matched-configuration gap is $+.112$ [$+.077, +.147$], $p = 5 \times 10^{-10}$, and a second seed of the whole pair gives $+.111$ (.937 vs. .826), with every cell within two points of seed 0. At 14B the gap is $+.078$ [$+.055, +.100$], $p = 7 \times 10^{-12}$, smaller because the facts-first model is at ceiling (.997). The two 14B checkpoints differ in the rest of the recipe bundle as well, so the 14B pair corroborates the direction at scale and the 1.7B pair is the single-variable test. The v2 releases carry the fix: on facts-first long states they score .947 (1.7B) and .950 (4B) against .598 and .612 for v1, and hold their facts-last accuracy (Table 21).

JevBench does not register the change (Appendix A.15): its 19-item `long_policy` family gives 6/19 against 2/19 ($p = .232$) at one seed and 5/19 against 5/19 at the other (Appendix A.11). The purpose-built 605-item set resolved at $p < 10^{-9}$ what a 19-item benchmark cell could not, and before the fix benchmark gains and the defect coexisted (Appendix A.6). The off-diagonal cells of Table 3 hold the evaluation render fixed; they measure a dependence on layout, which Section 6.3 takes up.

5.3 The rubric

The decision should change when the rubric changes. Workflow decisions arrive with a rubric that says how to decide, and nothing in ordinary training data forces a model to read it: when each state appears under a single rubric, the state and candidates can carry the answer on their own. A model never trained on the generator shows the cost on held-out families, grammars and rubric styles, where it scores .481/.507/.491, and on held-out rubric-flip pairs, where the same state and candidates take a different gold under a different rubric, it scores .477. All comparisons in this subsection and in Section 6.2 are matched 1.7B runs that differ only in data or sampler.

The fix is in the data. DecisionMix v2 (Section 3.3) builds every row into a counterfactual rubric group: the same state and candidate set under at least two rubrics with distinct gold answers. A predictor that ignores the rubric assigns one distribution to every member of a group, so on a two-member group its average cross-entropy stays at or above $\log 2$ nats, and the loss falls below that floor only by reading the rubric. Held-out compositions, domains, styles and grammars stay

evaluation-only, and level 7 (temporal, numeric, probability and trade-off composition) never enters training, so the test measures rubric reading on groups the model never saw (Appendix A.10).

Training on DecisionMix v2 lifts held-out families, grammars and rubric styles by 35–40 points over a model never trained on the generator (.481/.507/.491 $\rightarrow$.827/.881/.888, together with the uncertainty corpus), and accuracy on held-out rubric-flip pairs from .477 to .710; the model now depends on the rubric it is given (Appendix A.8). The shuffled-rubric control agrees: accuracy under a shuffled rubric falls from .446 to .405, so a wrong rubric now costs the model answers. The treatment also changed the sampler, the gains are measured within the generator’s own rule engine, level-7 composition moves only from .477 to .498 (Section 8), and the original workflow sets give up 3–5 points (Appendix A.8).

6 What the decision should not depend on

A decision should stay put when the order of the labels, the number of options offered or the layout of the input changes. Each subsection gives the failure a natural design shows, the perturbation that exposed it and the change that removed or bounded it.

6.1 Label order

The decision should not change when the same labels appear in a different order. A yes/no decision rendered as a two-way Choice fails this test. “Yes” and “no” occupy two different suffix positions with two different contextual states, so the rendering carries order into the scores, and reversing the labels of a two-way Choice over {no, yes} moves $P(\text{yes})$ by up to .55, and by .01–.15 on average across sets (Appendix A.7). Order invariance alone is easy to obtain: the candidate-blind variants of Section 5.1 are exactly set-stable and carry no question dependence (Appendix A.2).

Each output type delivers the property its contract names. Matched fine-tunes compare each type against a control that treats the same rows as plain Choice (Appendix A.7). The Noul Bernoulli head renders no candidates (Section 3.2), so label order cannot enter its computation. For rows routed to it, reversing the labels moves $P(\text{yes})$ by exactly zero on all nine test sets. The head is better or equal on six of seven yes/no sets and clears the untouched PagerDuty floor by a wide margin: accuracy .602 $\rightarrow$.886 against a .792 majority floor, NLL 1.04 $\rightarrow$.28. The contract holds only under per-row routing (Appendix A.7).

Choice has no output index for any label and must read the candidates to score them (Section 3.1), so the readout controls how much the order and the rest of the set move the scores (Table 2). N3 is far less sensitive to the candidate set than the letter-logit readout. Adding three unrelated options shifts N3’s pairwise log-odds by .09–.13 against .44–.46 for N1, below the teacher’s .179, and reordering moves its probabilities by .10–.11 against .165. The difference comes from the readout together with its null form; exact order invariance is the Bernoulli head’s contract.

6.2 How many options are offered

Whether the model abstains should depend on whether any offered candidate fits, and not on how many candidates are offered. Abstention belongs in the gate alone. A rendered “none of the above” line competes for softmax mass with the real candidates, so its share depends on how many there are. The K -sweep with the gold label absent exposes this. With the line, abstention has degenerate endpoints: the model almost never abstains on CLINC-150 at $K=5$ ($P(\emptyset \mid \text{absent}) = .02$) and always abstains at $K=150$ (1.00) and on Banking77 at $K=77$.

A matched pair tests the fix: both arms are N3 at layer 20 with the factored gate of Equation (2), and the treatment removes the rendered $\emptyset$ line. Removing the line removes both endpoints (null AUROC by K in Appendix A.2). MMLU-Pro false abstention falls from .692 to .003 (far outside its seed-to-seed variation of .05–.10) and accuracy rises from .133 to .353, Banking77 with all 77 labels rises from .063 to .579, and CLINC out-of-scope accuracy from .639 to .798, with Δ_q^{sh} unchanged (.118 vs. .122) and evidence tasks level (Table 8). The price is 2–7 points on intent and topic label spaces (HWU64 .801 $\rightarrow$.757, 20NG .589 $\rightarrow$.515) and a rise in reorder Δp from .077 to .123. This configuration, contextual readout at 71% depth with a gate and no rendered $\emptyset$ line, is the architecture of every checkpoint in this section and Section 7. The gate still has to learn when to fire, and the

training distribution sets that. Decision-type balance and null targets each move a separate axis from the rubric groups of Section 5.3. Duplicating 20% of workflow rows with the gold candidate removed and target $\emptyset$ lowers false abstention on CLINC-150 from .10 to .08 and on TREC-fine from .27 to .12, and raises accuracy by 2.6, 3.6, 4.3 and 6.0 points on CLINC-150, TREC-fine, HWU64 and 20NG (Table 10). The model now spends mass on $\emptyset$ where soft gold has none, so held-out Noul drops 4 points and held-out Score NLL rises ($1.52 \rightarrow 2.04$).

Family balance acts on false abstention as well. Adding workflow data at family weights $E .35 / K .25 / W .40$ cost 11 points on CLINC-150 and 10 on TREC-fine, almost all of it false abstention (CLINC-150 $.05 \rightarrow .17$). In the mixture sweep, which ran without the uncertainty corpus or null augmentation, $E \geq .45$ recovers CLINC-150 to .779 with false abstention .10 (Table 10). The release sampler, $E .40$ with $U .10$, keeps every evidence set within about 1 point of the evidence-only base (Appendix A.6).

6.3 Formatting

The decision should not change when the same content is laid out differently. Frozen models fail this test at 4B and 9B. With identical JevBench items, frozen Qwen3.5 checkpoints and letter-logit readout, a replication of another published system’s state and option layout lifts the standard tier by 1.4, 8.3 and 12.5 points at 2B, 4B and 9B, surviving a paired per-item test at 4B ($p = .048$) and 9B ($p < .001$); no hard-tier change separates (Table 9); the effect exceeds most of the spread between public JevBench entries (Appendix A.14).

A trained model depends on the render it was trained in. At 1.7B, render mismatch inside one trained model costs more than the training defect. In the off-diagonal cells of Table 3, switching the evaluation render costs 14.5 points for the facts-first model and 19.0 for the facts-last one, which lands on its policy_permit floor (.615 against .612). At 14B it costs 7.6 and 6.8 points, so the larger backbone tolerates render mismatch better.

Typical fixes the layout of the question and options in the suffix (Section 3.1), but the caller writes the state, and the render crossing of Section 5.2, which separates data effects from compatibility effects, measures what a change of state layout costs. The frozen controls of Table 4 use a letter-logit rendering we did not sweep, so frozen-versus-trained comparisons carry a rendering factor of the size shown here.

7 Speed, scale and probability quality

The architecture fixed in Sections 5 and 6 is the same at every backbone size. This section measures it against frozen backbones, in latency and in probability quality.

7.1 Scale

Every trained checkpoint beats its frozen backbone on the standard tier. Table 4 pairs each trained checkpoint with a frozen control: the same base model, untrained, reading next-token letter logits over the rendered options with three exemplars. The rows are not one recipe; Figure 3 plots them against backbone size.

The margins are 16.7, 2.8, 11.1 and 11.1 points at 1.7B (v1), 4B, 8B and 14B; each gap on its own sits inside its interval (paired test at 14B, frozen minus trained: $-.111 [-.250, +.014]$, $p = .10$). Question-conditioned knowledge grows with the backbone: Δ_q^{sh} rises from .127 at 1.7B to .235 at 14B, 2.3 times the 1.7B teacher’s; the 8B checkpoint is the exception (.093), and sits below the 4B and 14B under one matched recipe as well (Appendix A.2).

Qwen3.5-4B (typical-medium v2) is the strongest point per parameter: .861 standard and .495 hard, above every Qwen3 checkpoint we trained on the hard tier and 9.7 standard-tier points above its frozen control. Section 8 discusses the hard tier, where the frozen 14B leads.

7.2 Speed

Decision latency is nearly flat in backbone size. On the research harness a single decision at $K=2$ costs 45, 56 and 60 ms at 1.7B, 4B and 14B: an $8\times$ larger backbone is $1.3\times$ slower, because the

Table 4: Trained checkpoints with frozen controls. Recipe: *v1* = v1 release recipe (1,024-token window, no likelihood-based selection); *wf* = earlier workflow recipe (no DecisionMix, uncertainty corpus or typed heads); *fixed* = facts-first long states, truncated rows dropped, likelihood-based selection (+KD at 14B). JevBench (public subset); 95% cluster-bootstrap intervals (Section 4). MMLU-*K*: MMLU-Pro among-*K*. NLL: held-out Score. ms: research-harness single decision at *K*=2, one pod and build (the 8B was not on that pod). Checkpoints top to bottom: *ts1b*, *ts1c*, *tm1b*, *ladder_8b*, *t11b*, *tm2*.

Model	Recipe	JevBench		MMLU- <i>K</i>	Δ_q^{sh}	NLL	ms
		std [95% CI]	hard [95% CI]				
Qwen3-1.7B (small v1)	v1	.694 [.569, .819]	.432 [.342, .523]	.343	.127	1.01	45
small v2	fixed	.708 [.569, .833]	.432 [.342, .523]	.338	.127	1.08	–
frozen, 3-shot	–	.528 [.389, .667]	.369 [.279, .459]	–	–	–	–
Qwen3-4B (medium v1)	v1	.806 [.694, .903]	.423 [.333, .514]	.458	.193	1.03	56
frozen, 3-shot	–	.778 [.667, .889]	.441 [.351, .532]	–	–	–	–
Qwen3-8B	wf	.667 [.528, .792]	.396 [.306, .486]	.302	.093	1.67	–
frozen, 3-shot	–	.556 [.417, .694]	.369 [.279, .459]	–	–	–	–
Qwen3-14B	fixed	.931 [.861, .986]	.450 [.360, .541]	.498	.235	0.95	60
frozen, 3-shot	–	.819 [.708, .917]	.559 [.468, .649]	–	–	–	–
Qwen3.5-4B (medium v2)	fixed	.861 [.778, .944]	.495 [.405, .586]	.429	.194	0.97	–
frozen, 3-shot	–	.764 [.639, .875]	.495 [.405, .586]	–	–	–	–

cached state and a short suffix dominate (Appendix A.3). Cost stays flat to *K*=32 and rises at *K*=256, where the rendered candidates dominate the suffix (Section 3.1).

On the serving path a warm decision takes 15.5–17 ms at 1.7B and 19–21 ms for the Qwen3-4B model (warm p50, *K*=2, Figure 6; Qwen3.5-4B in Appendix A.4). Reading the candidates inside the forward pass (Section 5.1) has a price in throughput mode: with many questions against one cached state, a candidate-blind energy scorer costs about 1 ms per question at every *K* and is more than 3× cheaper than the contextual readout from *K*=32 on, measured on an earlier full-depth prototype (Appendix A.3).

7.3 Probability quality

Probability quality is indexed by decision type, so the training data, the target and checkpoint selection set it. The ordinal Score target shows the effect of the target alone. In the matched fine-tunes of Section 6.1, the ordinal-smoothed target ($\tau = 0.7$) adds no parameters and improves every probability-quality measure on held-out urgency at flat accuracy (.505 → .508): NLL 2.07 → 1.23, Brier .79 → .68, ECE .35 → .19, ordinal MAE .58 → .55. On JevBench’s 12 ordinal items the two models make identical per-item predictions, so the gain is in how mass is spread over adjacent levels.

A single post-hoc correction cannot do the same job: a global temperature and null-logit offset fit on held-out soft-target rows want opposite corrections for an evidence-and-knowledge and a workflow-trained checkpoint (Appendix A.5), so we make the correction in data and objective instead, through null augmentation and per-type targets.

On the 14B backbone the recipe bundle (facts-first long states with truncated rows dropped, a 3,072-token window, DecisionMix v2 and the uncertainty corpus, typed heads with the ordinal target, checkpoint selection on held-out uncertainty and curriculum NLL, and frozen-teacher distillation) takes held-out Score NLL from 2.87 under the earlier workflow recipe to 0.95, typed-decisions NLL from 1.96 to 1.04, and JevBench hard-tier Brier from .85 to .66 (Figure 7). No component is isolated except distillation, and the arm without it reaches the same Score NLL. The v1 1.7B and 4B checkpoints, which have the ordinal target and the uncertainty corpus but no likelihood-based selection, already reach 1.01 and 1.03; all 1.7B and 4B checkpoints with typed heads and DecisionMix sit at 0.97–1.08.

8 Limitations

Typical makes text-only decisions over candidate sets that fit in a short suffix; sets in the hundreds go through a shortlisting front end (Section 7.2). Multi-step composition is the main open problem: on DecisionMix level 7 (never trained) the v1 1.7B and 4B models score .495 and .544 against a .441 guess rate and the 14B model .620, and on the JevBench hard tier a frozen 14B backbone with three exemplars beats our trained 14B (paired $+ .108$ [$+ .027$, $+ .189$], $p = .015$; Appendix A.11). A fixed-depth pass has limited serial computation (Merrill and Sabharwal, 2023a), which intermediate tokens extend (Merrill and Sabharwal, 2023b; Feng et al., 2023). Formatting is the one factor in Table 1 that scale reduces without removing (6.8–7.6 points at 14B), and Choice is order-robust rather than order-invariant. The readout and evidence-first results are replicated across seeds; other rows are single runs, JevBench numbers use its 231 public items (unranked), and rubric-generalization results share one rule engine between training and held-out families. The training code is not yet released.

9 Conclusion

Typical reads decisions straight out of a pretrained language model. Given a document, a question and options supplied at request time, it returns a probability for each option and for abstaining, with no generated text, in 45 ms at 1.7B and 60 ms at 14B.

The central lesson is that such a model is only as good as what its decision depends on, and accuracy does not tell you. We tested each input and factor directly. In every configuration we tested, the decision carried question-dependent knowledge only when the candidates were read inside the forward pass: six designs that compute the decision state before seeing them all stayed within .03 of zero on Δ_q . It depends on the evidence when the head reads at 71% of depth, and on long documents when training places the evidence first (+11.2 points). It depends on the rubric when training uses counterfactual rubric groups (+35–40 points on held-out rule families). A Bernoulli head makes yes/no answers independent of label order, and removing the rendered “none of the above” option, leaving abstention to the gate, cuts MMLU-Pro false abstention from .692 to .003. Formatting is the one factor we reduced without removing: a trained model still depends on the layout it was trained in, less so at 14B.

The tests themselves, Δ_q and render-crossed evaluation, apply to any system that scores candidates, and we recommend reporting them alongside accuracy. The 1.7B and 4B models are public with an inference package. Multi-step composition is the open problem, and adding a small amount of intermediate computation before the typed readout is the natural next step.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016.
- Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. Artifacts or Abduction: How Do LLMs Answer Multiple-Choice Questions Without the Question? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10308–10330. Association for Computational Linguistics, 2024.
- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned Lens, 2023.
- Divyanshu Bhatt, Abhinav Joshi, Ujjaval Patel, and Ashutosh Modi. From representation to choice: Tracing decision emergence across languages in LLMs. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens, editors, *Findings of the Association for Computational Linguistics: ACL 2026*, pages 42423–42442, San Diego, California, United States, July 2026. Association for Computational Linguistics.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 632–642. Association for Computational Linguistics, 2015.
- Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325–331, December 2020.
- Iñigo Casanueva, Tadas Temčinas, Daniela Gerz, Matthew Henderson, and Ivan Vulić. Efficient Intent Detection with Dual Sentence Encoders. In *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, pages 38–45. Association for Computational Linguistics, 2020.

- Tongfei Chen, Zhengping Jiang, Adam Poliak, Keisuke Sakaguchi, and Benjamin Van Durme. Uncertain Natural Language Inference. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8772–8779. Association for Computational Linguistics, 2020.
- C. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1): 41–46, January 1970.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. BoolQ: Exploring the Surprising Difficulty of Natural Yes/No Questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 2924–2936. Association for Computational Linguistics, 2019.
- Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. Towards Revealing the Mystery behind Chain of Thought: A Theoretical Perspective, 2023.
- Eibe Frank and Mark Hall. *A Simple Approach to Ordinal Classification*, pages 145–156. Springer Berlin Heidelberg, 2001.
- Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks, 2017.
- In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt Cache: Modular Attention Reuse for Low-Latency Inference, 2023.
- Tilmann Gneiting and Adrian E Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477):359–378, March 2007.
- Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training Language Models With Pause Tokens, 2023.
- Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Gloriosi, and Daniel A. Roberts. The Unreasonable Ineffectiveness of the Deeper Layers, 2024.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks, 2017.
- Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training Large Language Models to Reason in a Continuous Latent Space, 2024.
- Dan Hendrycks and Kevin Gimpel. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. 2016.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring Massive Multitask Language Understanding, 2020.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network, 2015.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models, 2021.
- Samuel Humeau, Kurt Shuster, Marie-Anne Lachaux, and Jason Weston. Poly-encoders: Transformer Architectures and Pre-training Strategies for Fast and Accurate Multi-sentence Scoring, 2019.
- Daniel Keysers, Nathanael Schärli, Nathan Scales, Hylke Buisman, Daniel Furrer, Sergii Kashubin, Nikola Momchev, Danila Sinopalnikov, Lukasz Stafiniak, Tibor Tihon, Dmitry Tsarkov, Xiao Wang, Marc van Zee, and Olivier Bousquet. Measuring Compositional Generalization: A Comprehensive Method on Realistic Data, 2019.
- Omar Khattab and Matei Zaharia. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT, 2020.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles, SOSP '23*, pages 611–626. ACM, October 2023.
- Brenden M. Lake and Marco Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. 2017.
- Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. An Evaluation Dataset for Intent Classification and Out-of-Scope Prediction. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1311–1316. Association for Computational Linguistics, 2019.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast Inference from Transformers via Speculative Decoding, 2022.
- Zongxi Li, Xianming Li, Yuzhang Liu, Haoran Xie, Jing Li, Fu-lee Wang, Qing Li, and Xiaoqin Zhong. Label Supervised LLaMA Finetuning, 2023.

- Tom Lieberum, Matthew Rahtz, János Kramár, Neel Nanda, Geoffrey Irving, Rohin Shah, and Vladimir Mikulik. Does Circuit Analysis Interpretability Scale? Evidence from Multiple Choice Capabilities in Chinchilla, 2023.
- Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252. Association for Computational Linguistics, 2022.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.
- Xin Men, Mingyu Xu, Qingyu Zhang, Qianhao Yuan, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. ShortGPT: Layers in Large Language Models are More Redundant Than You Expect. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 20192–20204. Association for Computational Linguistics, 2025.
- William Merrill and Ashish Sabharwal. The Parallelism Tradeoff: Limitations of Log-Precision Transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023a.
- William Merrill and Ashish Sabharwal. The Expressive Power of Transformers with Chain of Thought, 2023b.
- Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. Adversarial NLI: A New Benchmark for Natural Language Understanding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4885–4901. Association for Computational Linguistics, 2020a.
- Yixin Nie, Xiang Zhou, and Mohit Bansal. What Can We Learn from Collective Human Opinions on Natural Language Inference Data? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9131–9143. Association for Computational Linguistics, 2020b.
- Rodrigo Nogueira and Kyunghyun Cho. Passage Re-ranking with BERT, 2019.
- Pouya Pezeshkpour and Estevam Hruschka. Large Language Models Sensitivity to The Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017. Association for Computational Linguistics, 2024.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3980–3990. Association for Computational Linguistics, 2019.
- Nithya Sambasivan, Shivani Kapania, Hannah Highfill, Diana Akrong, Praveen Paritosh, and Lora M Aroyo. “Everyone wants to do the model work, not the data work”: Data Cascades in High-Stakes AI. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI ’21, pages 1–15. ACM, May 2021.
- Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung-yi Lee, and Yun-Nung Chen. Let Me Speak Freely? A Study On The Impact Of Format Restrictions On Large Language Model Performance. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1218–1236. Association for Computational Linguistics, 2024.
- Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT Rediscovered the Classical NLP Pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601. Association for Computational Linguistics, 2019.
- Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, Paul Röttger, Frauke Kreuter, Dirk Hovy, and Barbara Plank. “My Answer is C”: First-Token Probabilities Do Not Match Text Answers in Instruction-Tuned Language Models. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 7407–7416. Association for Computational Linguistics, 2024a.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, pages 95266–95290. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024b.
- Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2547. Association for Computational Linguistics, 2025.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-Of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances*

- in *Neural Information Processing Systems 35*, NeurIPS 2022, pages 24824–24837. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2022.
- Adina Williams, Nikita Nangia, and Samuel Bowman. A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1112–1122. Association for Computational Linguistics, 2018.
- Hugh Mee Wong, Rick Nouwen, and Albert Gatt. When Models Decide and When They Bind: A Two-Stage Computation for Multiple-Choice Question Answering. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 22804–22821. Association for Computational Linguistics, 2026.
- Wenpeng Yin, Jamaal Hay, and Dan Roth. Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3912–3921. Association for Computational Linguistics, 2019.
- Urchade Zaratiana, Gil Pasternak, Oliver Boyd, George Hurn-Maloney, and Ash Lewis. GLiNER2: An Efficient Multi-Task Information Extraction System with Schema-Driven Interface, 2025.
- Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large Language Models Are Not Robust Multiple Choice Selectors, 2023.
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient Execution of Structured Language Model Programs. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, pages 62557–62583. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024.

A Appendix

A.1 Evaluation protocol details

Question-dependence gap. On one candidate-blind model the choices-only variant of Definition 4.1 reads $+0.017$ where Δ_q^{sh} reads $.000$, which is why the shuffled question is primary. High accuracy with $\Delta_q^{\text{sh}} \approx 0$ means that candidate-set priors explain the score. The reference is the list-wise teacher, a fine-tuned decoder reading the options in context. The zero band $|\Delta_q^{\text{sh}}| < .02$ is about 1.4 standard errors at $n=1,200$ ($p \approx .15$).

Evaluation sets. We evaluate on SNLI (Bowman et al., 2015), MNLI (Williams et al., 2018), ANLI (Nie et al., 2020a), BoolQ (Clark et al., 2019), CLINC-150 (Larson et al., 2019), TREC-fine, HWU64, 20 Newsgroups and Banking77 (Casanueva et al., 2020); a fixed 1,200-item MMLU-Pro probe (Wang et al., 2024b) and TruthfulQA-MC1 (Lin et al., 2022); held-out generator splits and ChaosNLI (Nie et al., 2020b); 605 held-out long states; and the evaluation-only external sets PagerDuty, jevlogs, Mind2Web, tree-choice, typed-decisions and JevBench. Each held-out split excludes a whole family, domain, rubric style, grammar or parameter range from training; none is a random validation split (Table 5). Banking77 is scored against all 77 labels. The MMLU-Pro probe is scored among the rendered options, with two leakage-matched items excluded. The held-out generator splits include the never-trained DecisionMix level 7 (880 items, guess rate .441). The 605 held-out long states split into `policy_permit` ($K=2$, $n=330$, floor .612) and `action_select` ($K=4$, $n=275$, floor .233). The evaluation-only external sets are PagerDuty (alert triage, yes/no, majority floor .792), jevlogs (log triage, floor .697), Mind2Web (web-agent actions, floor .427), tree-choice (taxonomy routing, $K \leq 320$), typed-decisions (400 cases of all three types, probabilistic gold) and JevBench.

JevBench. JevBench v1.2.1 has yes/no, choice ($K=2-6$) and ordinal decisions in three tiers: *standard* (72 items), *easy* (48) and *hard* (111). No training corpus contains a public id. Each item carries a state, a rubric and an exact label set. The hard tier spans eight families, including long policies, multi-hop, temporal, probability and trade-off. Its 146 judge-scored items are private and the harness ranks only submissions covering 95% of all items, which is why every number here is on the 231 public ids and unranked. The 72 standard items are 36 states in two paraphrases, so intervals are cluster bootstraps over the paraphrase group (20,000 resamples): $\pm 6-13$ points on standard, ± 9 on hard. The paraphrase pairs share a group field, over which the cluster bootstrap resamples. Comparisons between our models use paired per-item bootstraps.

Metrics. We report accuracy, Brier score, expected calibration error (ECE) and negative log-likelihood (NLL), with NLL and Brier primary on soft-gold sets. *Among- K accuracy* takes the argmax over candidates and ignores $P(\emptyset)$; the *false-abstain rate* is the fraction of gold-present items on which $\emptyset$ wins. *Rubric-flip* pairs share a state and candidate set under two rubrics with different gold. On MMLU-Pro, *reorder* Δp is the largest change in any candidate’s probability under reordering, and *pairwise* $\Delta \log\text{-odds}$ the shift in log-odds between the two top candidates when three unrelated options are added; both are zero for a set-invariant scorer.

Hardware and latency. Models train on single H100 GPUs (Appendix A.9). The *research harness* times one in-process decision on a 256-token state, state encode included, for K from 2 to 256, all sizes on one pod and PyTorch/CUDA build. The *serving path* is the released inference package, timed as warm p50 against an already-cached 256-token state at $K=2$.

A.2 Additional tables and figures

Table 5 lists the held-out axes of Appendix A.1. Tables 6 and 7 give the per-variant numbers for the candidate-blind states of Section 5.1, and Table 8 the rendered-null-line ablation of Section 6.2. Figure 3 plots Table 4 against backbone size. Under one matched recipe (the earlier workflow recipe run at 4B, 8B and 14B) the 8B is also below the 4B and 14B (.667 vs. .833/.875), as is its frozen control (.556 vs. .778/.819; Section 7.1). Table 9 gives the rendering-only comparison of Section 6.3. Figure 4 gives the long-state truncation histogram behind Section 3.3. The serving-path engineering behind Section 7.2, with Figure 6, is in Appendix A.4.

Table 5: Generalization axes, each with its own held-out split, and the scope a positive result licenses.

Axis	Evaluated on	A positive result licenses
Unseen examples	Held-out rows of every trained corpus	In-distribution competence
Unseen label word-ing	Label paraphrases; re-worded CLINC-150 labels	Robustness to label surface form
Unseen label space	Banking77-77, HWU64, tree-choice ($K \leq 320$), TruthfulQA-MC1	Scoring labels never trained on, within trained domains
Unseen family	3 of 15 workflow families; 3 Decision-Mix domains	Transfer across families of one generator
Unseen rubric style	2 withheld DecisionMix writing styles	Rubric conditioning beyond style memorization
Unseen grammar	1 withheld domain $\times$ level grammar	Rule execution inside the generator’s grammar
Unseen composition	DecisionMix level 7 (temporal, numeric, expected value, trade-off)	Composition outside the trained grammar
External	PagerDuty, jevlogs, Mind2Web, typed-decisions, JevBench	Transfer outside our generators

Table 6: Candidate-blind variants: six factorizations and objectives, two tap depths. “Shuffled” replaces the real question with an unrelated one, candidate set fixed; Δ_q^{sh} is the difference. $n=1,200$ MMLU-Pro items. Referenced from Section 5.1.

Variant	Factorization / objective	among- K	shuffled	Δ_q^{sh}
z1	single decision vector, bilinear scorer	.171	.147	+.023
zr	8 probes, token-level max-similarity	.157	.182	−.026
zr_set	+ $O(K)$ cross-attention over candidate tokens	.147	.151	−.003
ms_ctrl	single-set distillation, steps-matched control	.124	.147	−.023
ms_zr_set	same- Z multi-set, $\Delta\log$ -odds targets	.130	.130	+ .000
zr (full depth)	8 probes, tap at layer 28 of 28	.147	.156	−.008
Teacher	listwise decoder, options in context	.310	.208	+.102

Candidate-blind states learn calibration as well as priors: ECE on the knowledge probe falls from .39 for an earlier baseline to .04. The hypothesis-only SNLI baseline behind Section 5.2 scores 65.3 frozen and 68.2 with LoRA at layer 20; at layer 20 the N3 readout’s evidence accuracy trails the candidate-blind energy-scoring baseline most on ANLI, by 3.6. For the N3 readout, moving the tap from layer 28 to layer 20 also raises CLINC-150 from .740 to .862 and lowers validation NLL from .430 to .341 (Section 5.2). Decision features below the final, next-token-specialized layers agree with work on layer pruning and probing (Gromov et al., 2024; Men et al., 2025; Tenney et al., 2019; Belrose et al., 2023).

We compare the multi-set variant (ms_zr_set, Δ_q^{sh} exactly zero) against its steps-matched single-set control (ms_ctrl, −.023). Reading the full 28 layers lowers among- K accuracy to .147 and regresses SNLI/MNLI/ANLI/BoolQ to .881/.807/.462/.712 from .905/.873/.541/.826 at layer 20 (Table 7). The candidate-blind variants are exactly set-stable (reorder Δp and pairwise $\Delta\log$ -odds both zero).

No lever we varied moves Δ_q^{sh} (Section 5.1). Cross-attention from the probes over all candidate tokens leaves it at −.003 while improving null quality (CLINC out-of-scope .568 $\rightarrow$.727); supervising the change in the teacher’s log-odds across seven candidate sets gives exactly zero; reading all 28 layers gives −.008. Two variants fall outside the $\pm .02$ band in opposite directions (+.023, −.026), the spread 1,200 items allow. We cannot exclude optimization: the zero-initialized set-conditioning cross-attention stayed inert through 4k steps of direct supervision.

With the rendered $\emptyset$ line, null AUROC is .97/.94/.91 at $K = 5/20/50$. Without it, null AUROC is .98/.95/.92/.95 on CLINC-150 at $K = 5/20/50/150$ and .90/.83/.69 on Banking77 at $K = 5/20/77$ (Section 6.2). Removing the line also takes the KL divergence to the teacher on MMLU-Pro from 2.43 to .27. It costs HWU64 .801 $\rightarrow$.757 and 20NG .589 $\rightarrow$.515, and raises reorder Δp from .077 to .123.

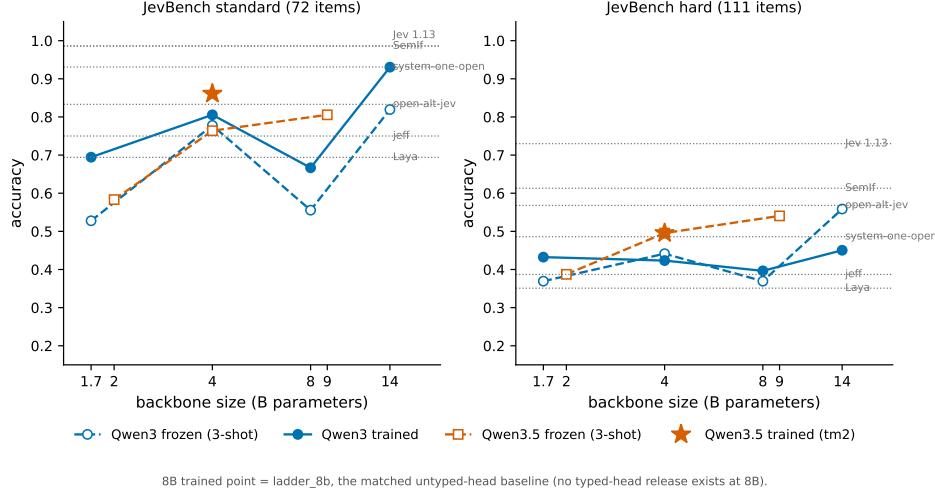


Figure 3: JevBench (public subset) standard and hard accuracy against backbone size. Qwen3 trained (solid; each point uses the recipe named for its row in Table 4) and frozen with three exemplars (dashed); Qwen3.5 frozen controls at 2B/4B/9B and the trained Qwen3.5-4B (star). Dotted lines mark other public entries on the same ids, whose rendering and protocols differ (Appendix A.14). Intervals are in Table 4.

Table 7: Candidate-blind Z , closure at full depth vs. the original tap-20 student.

	Tap 20	Full depth (28)	Teacher
MMLU-Pro among- K / choices-only / shuffled-q	.157 / .166 / .182	.147 / .177 / .156	.310 / .222 / .208
Δ_q (choices-only) / Δ_q^{sh} (primary)	-.009 / -.026	-.030 / -.008	.088 / .102
SNLI / MNLI / ANLI / BoolQ	.905/.873/.541/.826	.881/.807/.462/.712	-

A.3 Native choice vs. energy scoring vs. batched log-probability, full K -sweep

On the research harness the single-decision cost stays flat to $K=32$ (46, 56 and 62 ms at 1.7B, 4B and 14B) and reaches 106, 118 and 157 ms at $K=256$ (Table 26). With $M=32$ questions against one cached state, the batched marginal cost per question is 2.7 and 3.9 ms at $K=2$ for 1.7B and 14B, and 28.0 and 109.7 ms at $K=256$. In that throughput mode a candidate-blind energy scorer costs about 1 ms per question at every K and is more than $3\times$ cheaper than the contextual readout from $K=32$ on, measured on an earlier full-depth prototype (below).

Throughput mode on an earlier full-depth prototype. These numbers come from a different architecture and harness than Section 7.2: the full-depth (28-layer) contextual-readout prototype of Section 5.1, timed before the one-pod comparison. At $M=256$ queries against one cached 256-token state, its per-query marginal cost is 1.07, 1.55, 3.38 and 43.8 ms at $K=2, 10, 32, 256$; a candidate-blind energy-scoring path costs 0.97–1.06 ms across the same range (the contextual readout is $1.1\times$, $1.6\times$, $3.4\times$ and $41\times$ its cost); and a baseline that re-runs the backbone once per candidate costs 2.1, 4.6, 12.2 and 101.7 ms. We transcribed these values from the run’s log after an out-of-memory crash lost the run’s bench.json. We measured timing only and did not evaluate a shortlist-then-decide pipeline for accuracy. On the one-pod harness, batched marginal cost at $M=32$ scales more steeply with both size and K than single-decision cost, from 2.7 ms at 1.7B/ $K=2$ to 109.7 ms at 14B/ $K=256$.

A.4 Serving-path engineering

The first version of the public inference package deep-copied the entire cached prefix’s key-value tensors on every decision. We replaced the copy with a zero-copy strided view plus cached attention masks and position tensors, which cut warm per-decision latency from 21–27 ms to 15.5–17 ms (typical-small v1, Qwen3-1.7B) and 19–21 ms (typical-medium v1, Qwen3-4B) on one H100

Table 8: Rendered $\emptyset$ line vs. the factored null, matched checkpoints (tap 20).

	Rendered $\emptyset$ line	Factored null (no rendered line)
MMLU-Pro false-abstain rate	.692	.003
MMLU-Pro accuracy	.133	.353
Banking77 (unseen labels) accuracy / false-abstain	.063 / .924	.579 / .135
CLINC-OOS accuracy / null AUROC	.639 / .840	.798 / .973
Δ_q^{sh} (unchanged)	.118	.122

Table 9: Rendering alone, frozen weights. Identical Qwen3.5 base checkpoints and items, JevBench (public subset), letter-logit readout in both columns; only the state and option rendering differs. Δ : paired cluster bootstrap over the per-item difference, 20,000 resamples. Referenced from Section 6.3.

Tier	Backbone	our rendering	replicated rendering	paired Δ [95% CI], p
standard	Qwen3.5-2B	.583	.597	+.014 [−.194, +.208], $p=.95$
	Qwen3.5-4B	.764	.847	+.083 [+ .014, +.167], $p=.048$
	Qwen3.5-9B	.806	.931	+.125 [+ .056, +.208], $p<.001$
hard	Qwen3.5-2B	.387	.441	+.054 [−.045, +.153], $p=.33$
	Qwen3.5-4B	.495	.468	−.027 [−.117, +.063], $p=.64$
	Qwen3.5-9B	.541	.595	+.054 [−.027, +.135], $p=.22$

(Figure 6). tm2 (Qwen3.5-4B, typical-medium v2) serves at 34–46 ms warm p50 on the same package with reference DeltaNet kernels. torch.compile/CUDA-graph capture reached 8 ms at a single fixed input shape, but probabilities shifted by up to .1 across the shape buckets the serving path sees, and the mutable key-value cache produced stale-buffer crashes under the standard capture-and-replay pattern, so we reverted it. Manual per-bucket graph capture is the remaining path toward ~ 10 ms decisions.

A.5 Probability quality: global fit and NLL by checkpoint

Figure 7 gives held-out Score and typed-decisions NLL by checkpoint for Section 7.3. Under the earlier workflow recipe, with selection on in-distribution validation loss, held-out Score NLL was 2.03, 2.15, 1.67 and 2.87 at 1.7B, 4B, 8B and 14B.

Fitting one global temperature and null-logit offset jointly on held-out soft-target rows gives opposite corrections for two checkpoints (Section 7.3). For an evidence-and-knowledge checkpoint the fitted offset is -1.75 , which cuts CLINC-150 false abstention from .053 to .022. For a workflow-trained checkpoint it is $+3.0$, which repairs typed-decisions NLL ($1.95 \rightarrow 1.19$) and held-out Score NLL ($2.03 \rightarrow 1.09$) and collapses intent and topic coverage to 5–9%. We make the correction in data and objective instead, through null augmentation and per-type targets (Sections 5.3 and 6.1).

A.6 Mixture sweep, full table

Table 10: Mixture sweep at 1.7B, matched steps, referenced from Section 5.3. “6A” is the $E:0.35/K:0.25/W:0.40$ point; “e45+null” adds null_aug W:0.20 to $E:0.45/K:0.20/W:0.35$.

	Base (E only)	6A	e45	e45+null	long _{e45} (1024 tok)
CLINC-150 acc / false-abstain	.845/.05	.734/.17	.779/.10	.805/.08	.741/.13
MMLU-Pro among- K / Δ_q^{sh}	.353/.122	.330/.119	.338/.115	.353/.143	.329/.116
Held-out score NLL	—	2.03	1.52	2.04	1.26
JevBench std / hard	.694/.378	.750/.387	.833/.396	.806/.396	.806/.450

In this sweep, which ran without the uncertainty corpus or null augmentation, $E \geq .45$ brings NLI and BoolQ back within 1 point of the evidence-only base and intent and topic sets within 3–4, with workflow metrics within 1–4 (Section 5.3); CLINC-150 recovers to .779 with false abstention .10, and to .805 and .08 once null augmentation is added. With null augmentation, MMLU-Pro among-

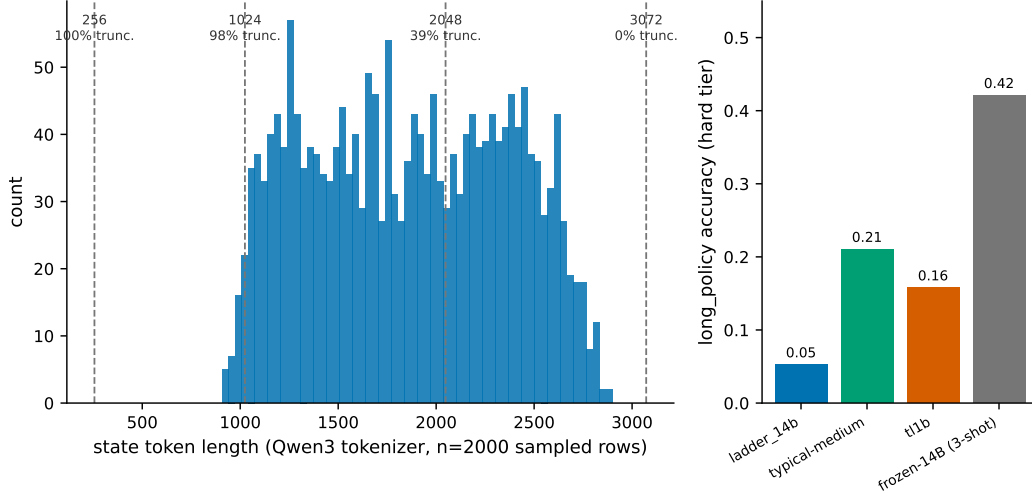


Figure 4: Left: histogram of `data_wf_long` state token lengths with truncation fractions at 256/1024/2048/3072 tokens. Right: `long_policy` (hard-tier) accuracy across the long-state bug/fix sequence.

K returns to the evidence-only base (.353), with the highest Δ_q^{sh} of any mixture (.143). Before the facts-first fix (Section 3.3), the long-state row (long_{e45} , 1,024 tokens, 25% long-policy rows) alone raised `long_policy` from .16 to .37 and `multi_hop` from .17 to .28 relative to 6A, and gave the best held-out Score NLL of the sweep (1.26). Benchmark gains and the long-state defect therefore coexisted (Section 5.2). Null-augmentation fractions in [.10, .20] give the same model within noise.

A.7 Typed primitives: parameterization, routing, and full results

Choice is the K -way readout of Section 3.1, unchanged: a hard one-of- K target trained with soft cross-entropy against Equation (2). Score is ordinal (severity, quality, urgency). Plain Choice penalizes a one-level miss as much as an opposite-end miss, so Score uses an ordinal-smoothed target,

$$\tilde{t}_j \propto \exp\left(-\frac{|j - y|}{\tau}\right), \quad \tau = 0.7, \quad (3)$$

normalized over valid levels, with $\tau=0$ recovering one-hot Choice. Only the target changes; the architecture stays the same. Noul is a yes/no proposition. Rendered as 2-way Choice it has no order-invariance guarantee, since “yes” and “no” occupy two different suffix positions with two different contextual states. We therefore give Noul a dedicated Bernoulli head that reads the query alone, with no candidate text,

$$P(\text{yes}) = \sigma(w_n \cdot h_D), \quad (4)$$

which is exactly order-invariant by construction for rows routed to it; the reversed-label control confirms this with an exact zero. The remaining mass $1 - P(\text{yes})$ goes to the other candidate. The log of these two probabilities takes the place of the candidate scores in Equation (2), and a separate gate reading h_D supplies $P(\emptyset)$, so the output keeps the factored abstention form.

Per-row routing. A row is Noul-eligible iff its candidate set is exactly {yes, no}; every other row scores as ordinary Choice, bit-identical to a run with Noul disabled. The type is a contract on the input. An early release candidate broke that contract by applying the Bernoulli head at the *model* level, so it scored Choice- and Score-typed rows in the same batch with no candidate text, and its accuracy collapsed to chance. Routing per row, which 140 unit tests verify, restored it.

The matched fine-tunes of Section 6.1 start from one parent checkpoint, and each trains for 3k steps on the rows of its own type.

Reversing the labels of a two-way Choice moves $P(\text{yes})$ by .01–.15 on average across sets, and the released v1 checkpoint shows zero on 10 of 11 sets and .009 on one. On the held-out hard split of a trained typed-decision source (`systemone-lite`) the ordinal-smoothed target also lifts accuracy from .853 to .939 and cuts MAE from .15 to .06 (Section 6.1).

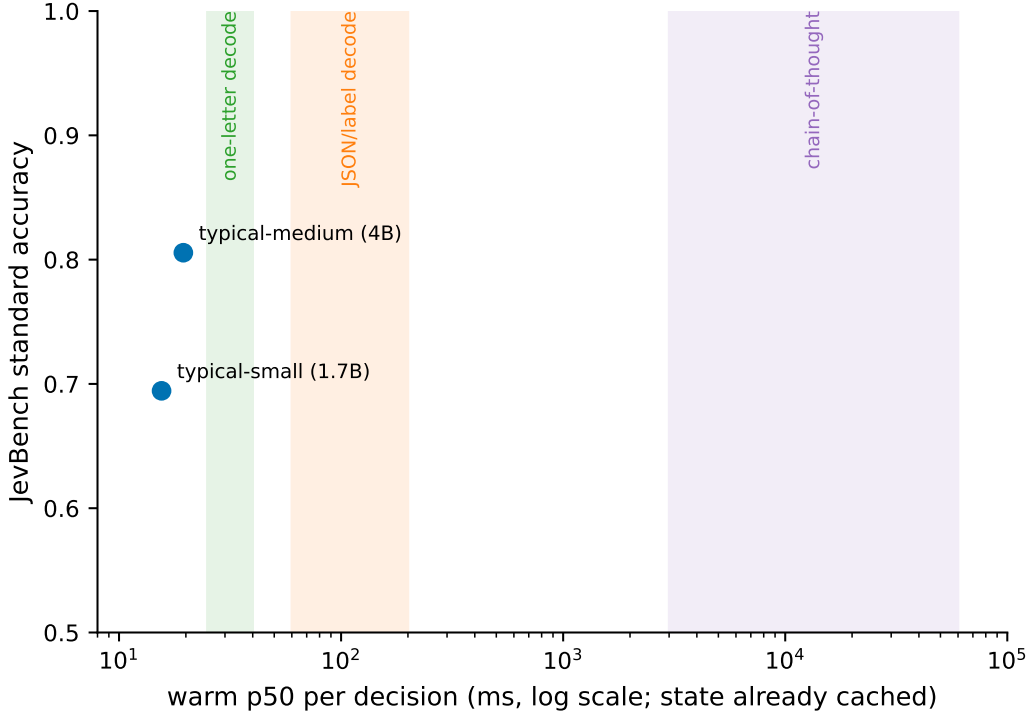


Figure 5: JevBench (public subset) standard accuracy vs. *warm* per-decision p50 latency (ms, log scale) for the v1 release checkpoints (ts1b, tm1b), with generative-LLM decode-latency comparison bands. Latency is the served path with the state prefix already cached (runs/serve_bench2, $K=2$, 256-token state, one stream, in process, model load excluded), the same quantity as Appendix A.4. The single-decision ladder of Table 26 measures a different quantity: it includes the state encode on the unoptimized path. Referenced from Section 7.2.

Table 11: Typed primitives vs. their K -way Choice control, matched fine-tunes, referenced from Section 6.1.

Score	K -way Choice	Ordinal-smoothed ($\tau=0.7$)
Held-out urgency: acc / NLL / ECE	.505 / 2.07 / .35	.508 / 1.23 / .19
JevBench hard: acc / Brier	.360 / .88	.459 / .70
JevBench ordinal items (12), decisions changed	—	0 of 12
Noul	2-way Choice	Bernoulli
PagerDuty (floor .792): acc / NLL	.602 / 1.04	.886 / .28
Reversed-label max $ \Delta P(\text{yes}) $	.55	0 (exact, all 9 sets)

A.8 DecisionMix v2, full numbers

Held-out family/grammar/style accuracy moves from .48/.51/.49 (control, never trained on data_wh or data_u) to .83/.88/.89 (DecisionMix v2 together with the uncertainty corpus; the treatment also changed the sampler), and rubric-flip accuracy moves from .48 to .71, measured within the generator’s own rule engine (Section 3.3). Accuracy under a shuffled rubric falls from .446 to .405. Level-7 composition, which the generator never produces, moves from .477 to .498 (Section 8). The uncertainty corpus moves typed-decisions NLL from 2.06 to 1.71 and ChaosNLI NLL from 1.27 to 1.00 while ChaosNLI accuracy falls from .584 to .537 (Section 3.3), trading top-1 accuracy for likelihood. The original workflow sets give up 3–5 points at a fixed workflow share now split over four corpora (Section 5.3). Level-7 composition is in Appendix A.11.

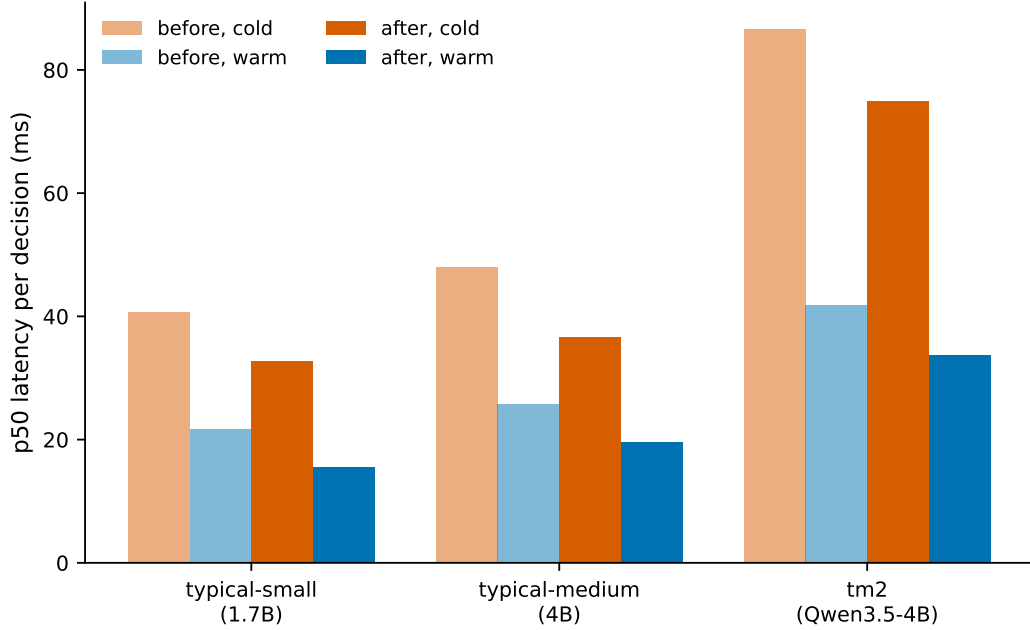


Figure 6: Per-decision p50 latency before/after the zero-copy prefix fix, cold vs. warm, for typical-small v1 (1.7B), typical-medium v1 (Qwen3-4B), and tm2 (Qwen3.5-4B, typical-medium v2), at $K=2$, 256-token state.

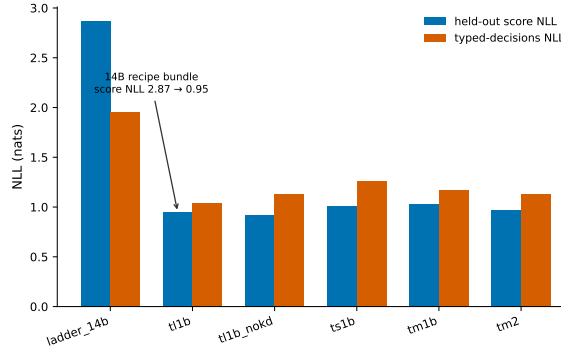


Figure 7: Held-out Score and typed-decisions NLL (held-out sets, not JevBench). ladder_14b: the 14B under the earlier workflow recipe; t11b: the same backbone under the recipe bundle of Section 7.3; t11b_nokd: the bundle without distillation. ts1b, tm1b, tm2: 1.7B and 4B checkpoints with typed heads and DecisionMix.

A.9 Full hyperparameters

Table 12 gives the exact configuration of each checkpoint in the paper. All runs use AdamW, bf16, and LoRA rank 16 on the top eight kept layers of the truncated backbone. Section 3.3 describe the family-balanced sampler and null augmentation.

ts1b/tm1b also set null_aug W:0.20, ordinal_smooth 0.7 (Score), noul_head bern (routed per row, Section 3.2), null factored, and nc_render letters_nonull. t11b adds -drop_truncated, frozen-teacher KD ($\alpha \cdot \text{CE} + \beta \cdot \text{KL}$, $T=2$, against the same backbone’s own frozen 3-shot letter distribution on 63k labelled rows), and checkpoint selection on held-out uncertainty-plus-curriculum validation NLL (-best_on) in place of in-distribution loss. For ts1b we fit the final temperature and null offset post hoc, $T=1.124$ and offset 0, and apply them at evaluation. ts1c (typical-small v2) is the 1.7B model trained on the facts-first long-state corpus with

Table 12: Training configuration by checkpoint. Tap is given as layer / total layers of the base model. $E:K:W:U$ is the family-sampler weight per batch. Max state is the training-time token budget for the cached prefix.

Checkpoint	Backbone	Tap	Steps	Batch	$E:K:W:U$	Max state
typical-small-preview	Qwen3-1.7B-Base	20/28	12,000	64	.35:.25:.40:0	256
ts1b (typical-small v1)	Qwen3-1.7B-Base	20/28	12,000	64	.40:.15:.35:.10	1,024
tm1b (typical-medium v1)	Qwen3-4B-Base	26/36	12,000	64	.40:.15:.35:.10	1,024
ladder_8b	Qwen3-8B-Base	26/36	12,000	64	.35:.25:.40:0	256
ladder_14b	Qwen3-14B-Base	28/40	12,000	64	.35:.25:.40:0	256
tl1b (14B)	Qwen3-14B-Base	28/40	8,000	64	DecisionMix v2 + U , facts-first	3,072
tm2 (typical-medium v2)	Qwen3.5-4B-Base	$\approx 71\%$	12,000	64	.40:.15:.35:.10	2,048

a 2,048-token window, `-drop_truncated` and the same `-best_on` selection for 8,000 steps; its long-state and JevBench numbers are in Table 21. tm2 also uses `-best_on`; ts1b and tm1b have no likelihood-based selection.

A.10 Corpus construction detail

`data_v5` (633,213 train / 4,001 val rows) covers evidence tasks and classification with a wide span of label vocabularies at low rows-per-source, so the model cannot memorize a fixed small set of label sets and must read the rendered options.

`data_kb` (147,447 train / 3,001 val) is distilled from ARC-Easy/Challenge, OpenBookQA, CommonsenseQA, SciQ, QASC, LogiQA, AQuA-RAT, MedMCQA, and MMLU-auxiliary-train. We hold out TruthfulQA-MC1, and after a leakage audit against MMLU-Pro we excluded two items from all MMLU-Pro numbers.

`data_wf` (72,006 train / 3,000 val) is our own rubric-conditioned workflow generator, 15 families of Noul/Choice/Score decisions (policy checks, adequacy, support, fact verification, eligibility, routing, action selection, enumeration extraction, categorical judgment, tool selection, and four ordinal quality/severity/relevance/completeness scales, plus urgency). We hold three whole families (eligibility, tool_select, urgency, one per type) out of training. Of the 72,006 rows, 26,513 belong to *rubric groups*: the same state and candidate set rendered under two or three different rubrics with different gold answers. We built these groups to measure and train rubric dependence over state/candidate pattern-matching.

`data_wf_hf` (86,600 train rows) adds three externally sourced sets in the same typed shape: `cua-s1-forms` (60,000 rows, procedural GUI-form option-scoring, MIT license, with a form-signature-disjoint 24,370-row test split and a 196-row real-usage demo split held out), `systemone-lite-general` (20,000 rows, rule-labeled typed decisions, MIT, with a 5,400-row hard split held out), and `jev-4b-distill-data` (6,600 rows, programmatic gold, Apache-2.0, with a 600-row adversarial split held out).

`data_wf_long` (23,318 train rows) is a longer-state variant of the workflow generator for multi-paragraph policy documents, rendered facts-first (Section 3.3).

`data_wh` (60,605 train / 3,000 val) is a second, independent rubric-conditioned generator (Decision-Mix v2): a programmatic rule engine over 12 domains and six boolean conditions at seven levels of compositional depth. We train on levels 1–6 and keep level 7 (compositions requiring temporal, unit-conversion, expected-value, or trade-off reasoning that the generator does not otherwise produce) for evaluation only. Every trained row belongs to a counterfactual rubric group. We hold out one whole domain/level pair, three whole domains, and two rubric styles.

`data_u` (31,029 train / 2,000 val) is a soft-target uncertainty corpus: the validation split of UNLI, the ambiguous rows of AmbiEnt (uniform target over the listed labels), and four synthetic generators with exact closed-form posteriors (partial-evidence combination, a noisy-sensor Bayesian update, ordinal confusion via an inverted confusion matrix, and conflicting-sources log-odds pooling), each with a held-out parameter range; ChaosNLI-ambiguity (MNLi portion) is evaluation-only.

Table 13: data_wf: 15 families, 72,006 train rows, 3 held out (marked *). “In rubric groups” counts rows that are part of a same-state-same-candidates, different-rubric-different-gold group.

Family	Type	Rows	In rubric groups
policy_permit	Noul	9,000	5,205
adequacy	Noul	5,000	0
support	Noul	6,000	0
fact	Noul	5,000	0
eligibility*	Noul	1,500	851
routing	Choice	9,003	5,208
action_select	Choice	7,000	4,054
enum_extract	Choice	7,000	3,618
categorical	Choice	7,004	3,882
tool_select*	Choice	1,500	905
quality	Score	6,000	0
severity	Score	6,000	3,497
relevance	Score	3,998	0
completeness	Score	4,001	2,196
urgency*	Score	1,502	900

Table 14: data_wh (DecisionMix v2): 60,605 train rows across 12 domains $\times$ 6 boolean conditions, levels 1–6; level 7 (temporal/numeric/probability/trade-off composition) is eval-only.

Level	Train rows	Held-out grammar rows	Eval file	Rows
1	10,601	142	wh_heldout_family	422
2	10,600	141	wh_heldout_style	403
3	10,600	141	wh_heldout_grammar	816
4	10,601	140	wh_level7	880
5	10,602	142	wh_rubric_flip	2,206
6	10,601	141	wh_rubric_shuffled	2,437

A.11 Hard compositional tier and the frozen 14B comparison

Single-pass readout reaches its frontier on compositions deeper than anything in training. Rubric-conditioned training moves held-out families, grammars and styles inside the generator’s distribution to .84–.90 (Appendix A.8). Level-7 composition, which requires temporal arithmetic, unit conversion, expected-value reasoning or multi-attribute trade-offs, moves far less. On the 880-item level-7 set the uniform-guess rate is .441 (598 items at $K=2$, 224 at 3, 58 at 4; the majority-class rate is .250), and the checkpoints score .495 (ts1b), .544 (tm1b) and .620 (t11b). Only the 14B checkpoint sits well above the guess rate, and all three sit 25–40 points below their own in-distribution held-out scores: the curriculum that moves every in-distribution family by more than 30 points moves level 7 by 2 (.477 to .498). The JevBench hard tier’s temporal, probability and trade-off families show the same pattern.

The frozen 14B backbone with three exemplars leads the trained 14B on the hard tier. Table 16 compares the previous-recipe 14B checkpoint (ladder_14b); the bundle that fixed the long-state rendering, dropped over-length rows, and added DecisionMix v2, the typed heads and likelihood-based checkpoint selection (t11b); the same bundle without frozen-teacher distillation (t11b_nokd); and the frozen backbone with three exemplars. A paired cluster bootstrap on the per-item difference over the same 111 hard items gives frozen-minus-trained $+.108$ [$+.027$, $+.189$], $p=.015$ against t11b, and $+.081$ [$-.009$, $+.171$], $p=.082$ against t11b_nokd. On the standard tier the ordering reverses (frozen-minus-trained $-.111$ [$-.250$, $+.014$], $p=.10$). We scored the frozen control with a 3-shot letter-logit rendering that we did not sweep; rendering alone moves a frozen 9B checkpoint by 12.5 standard-tier points (Table 9).

The backbone carries hard-tier behavior that the current recipe does not preserve. Per family (Figure 8, Table 18), the frozen 14B leads t11b on tradeoff (.500 vs. .167), long_policy (.421 vs. .158), ambiguous (.714 vs. .429), probability (.500 vs. .300) and temporal_numeric (.333 vs. .200), and trails on multi_hop (.444 vs. .611). Families hold 3–19 items and no single family separates; the direction is the same across every serial family. Our generators compose boolean

Table 15: data_u: 31,029 train rows, 100% soft-target.

Source	Train rows
Synthetic (4 closed-form generators)	28,560
UNLI (validation split only)	2,068
AmbiEnt (ambiguous rows)	401
By decision type: Noul / Score / Choice	23,403 / 7,225 / 401

Table 16: 14B comparisons. JevBench (public subset); 95% cluster-bootstrap intervals over the paraphrase group field ($n_{\text{eff}}=36$ standard states, $n=111$ hard items). t11b_nokd is identical to t11b in every flag except -distill_beta 0. Paired per-item tests are in the text; the marginal intervals overlap, and no comparison rests on them.

	ladder_14b	t11b	t11b_nokd	Frozen, 3-shot
JevBench std	.875 [.792, .944]	.931 [.861, .986]	.917 [.833, .986]	.819 [.708, .917]
JevBench hard	.468 [.378, .559]	.450 [.360, .541]	.477 [.387, .568]	.559 [.468, .649]
JevBench Brier std (hard)	.173 (.85)	.175 (.66)	.127 (-)	.30 (.60)
long_policy (hard subfamily)	.053	.158	.211	.421
Training val NLL	-	.438	.410	-
Held-out score NLL	2.87	0.95	-	-
Level-7 composition (guess rate .441)	-	.620	-	-

rules up to six levels deep and never produce the temporal, probability and trade-off operations these families use, which explains why training did not add the behavior. The measurement that separates missing coverage from a limit of fixed-depth computation is the same frozen backbone given a scratchpad on the same 111 items; Section 9 takes this up.

Same-backbone frozen-teacher distillation has no detectable effect. It is the one component of the 14B bundle with its own matched control. t11b_nokd has the better point estimate on hard accuracy (.477 vs. .450), long_policy (.211 vs. .158), training validation NLL (.410 vs. .438) and standard-tier Brier (.127 vs. .175); t11b has the better one on standard accuracy (.931 vs. .917). The paired bootstraps are $-.027 [-.090, +.027]$ ($p=.45$) on hard and $+.014 [-.042, +.069]$ ($p=.81$) on standard.

The bundle’s clear effect is on probability quality. The ladder_14b $\rightarrow$ t11b standard-tier change is $.875 \rightarrow .931$ (paired $+.056 [+.000, +.125]$, $p=.13$). Held-out score NLL falls from 2.87 to 0.95 and hard-tier Brier from .85 to .66, both large relative to their noise.

The same question on 19 JevBench items. ladder_14b scores .053 on the JevBench long_policy family ($n=19$) and .919 on the 605 held-out long states in its own training render (Table 3), a pair that also changes the item set and stacks truncation damage, render mismatch and a 19-item sample. Before the 2×2 of Section 5.2, we asked the facts-first vs. facts-last question on JevBench (Table 17). The two 1.7B arms are identical in every flag except whether data_wf_long renders its facts at 0% or 97.6% of the state text (same rows, same labels, matched state length, $p50 = 8,702$ characters in both arms), trained at -max_state 1024 with -drop_truncated off so that truncation bites exactly as it did before the fix. On 19 long_policy items in one fixed benchmark render the seed-0 arms read .316 (6/19) vs. .105 (2/19), Fisher exact $p=.232$, with the hard-tier aggregate moving the other way. A second seed of the whole pair, byte-identical flags apart from the seed, reads 5/19 vs. 5/19: the 19-item family gives a four-item gap at one seed and none at the other. On the 605 held-out long states the same pairs give matched-configuration gaps of $+.112 [+.077, +.147]$, $p = 5 \times 10^{-10}$ (seed 0) and $+.111$ (seed 1, .937 vs. .826), and the 14B pair gives $+.078 [+.055, +.100]$, $p = 7 \times 10^{-12}$ (Table 3).

A.11.1 Per-family JevBench-hard table

Table 18 gives the hard-tier sub-families at 14B. We read the frozen-control column and the ambiguous row from each run’s JSON evaluation artifact. For ladder_14b’s temporal and probability cells, the JSON artifact and the project’s internal report disagree (.333/.50 in the JSON vs. .20/.40 in the report); the table gives the report values, marked †.

Table 17: Matched facts-first vs. facts-last training arms, 1.7B, single variable, two seeds, scored on JevBench: the question of Table 3 asked on 19 long_policy items in a single fixed render. The last row gives the same pairs on the 605 held-out long states. JevBench (public subset), 95% cluster-bootstrap intervals.

	Arm A (facts-first)	Arm B (facts-last)	Test
<i>seed 0</i>			
long_policy	.316 (6/19)	.105 (2/19)	Fisher $p=.232$; bootstrap Δ [$-.053, +.474$]
JevBench hard	.378 [.288, .468]	.396 [.306, .486]	opposite direction
JevBench std	.750 [.625, .861]	.736 [.597, .861]	indistinguishable
<i>seed 1</i>			
long_policy	.263 (5/19)	.263 (5/19)	tie
JevBench hard	.405	.387	–
JevBench std	.792	.708	–
605 states, matched	.942 / .937	.830 / .826	gap $+.112$ (s0), $+.111$ (s1)

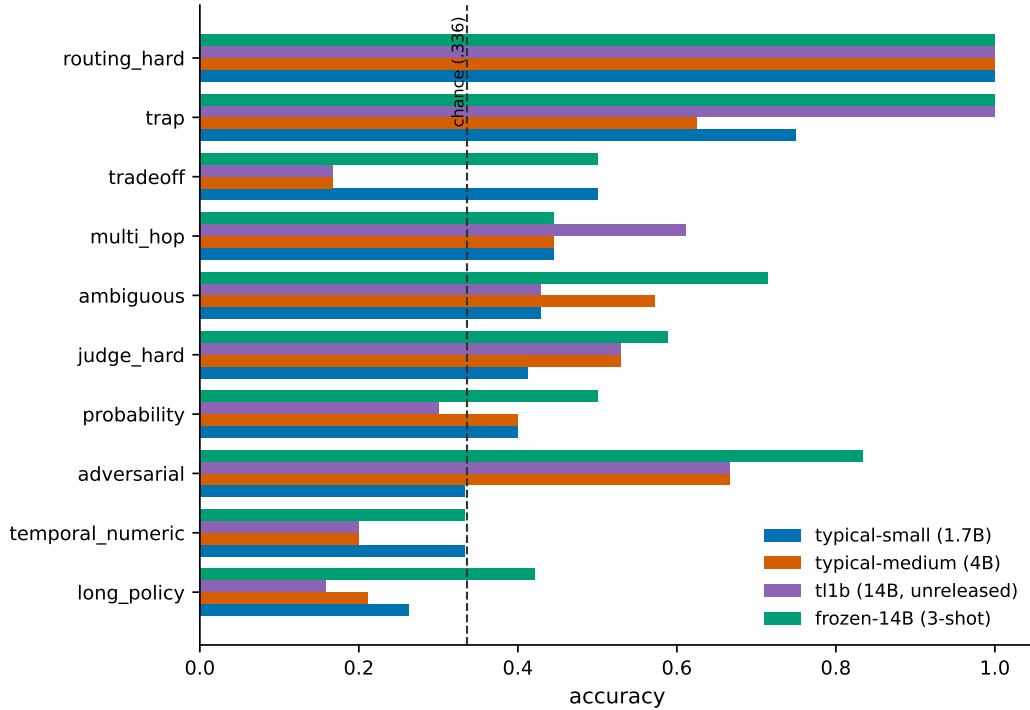


Figure 8: Per-family JevBench (public subset) hard-tier accuracy for the v1 release checkpoints (ts1b, tm1b), the 14B checkpoint t11b, and the frozen-14B 3-shot control, chance line at .336. Per-family n ranges from 4 to 19 items; no single family separates, and the direction across the serial families is uniform. Values come from each run’s JSON evaluation artifact.

A.12 Closed-loop decisions: a case study

We drove three games with typical-small through the public decision API, one decision per tick, on a laptop (Apple M4 Pro, MPS; the probes ran against the v1 weights, before the v2 upload). Each tick the game engine renders the current situation as the state, offers the legal actions as the candidate set, and executes the argmax. The state holds only the situations that apply at that tick, each written as a sentence that already contains the comparison a rule needs. Drive and Doom put a precedence-ordered list of one-condition rules in the question; Snake encodes its rules in the state sentences and asks a plain question. Accuracy is agreement with the literal rule list applied over the offered actions (Table 19).

Table 18: JevBench (public subset) hard-tier sub-family accuracy, 14B; per-family cells are 3–19 items each and carry no useful interval.

Sub-family	ladder_14b	t11b	Frozen 14B, 3-shot
long_policy	.05	.158	.421
multi_hop	.44	.611	.444
trap	1.00	1.00	1.00
adversarial	.67	.667	.833
ambiguous	.429	.429	.714
temporal	.20 [†]	.20	.333
probability	.40 [†]	.30	.50
tradeoff	.67	.17	.50

[†] The JSON-derived values for these two ladder_14b cells are .333 (temporal) and .50 (probability).

Table 19: Closed-loop decisions with typical-small. “Probe” is agreement with the literal rule list on scripted situations; “chance” is uniform over the offered legal actions; “no rules” keeps the state and replaces the rule list with a goal question.

Game	Candidates (legal ac- Probe Chance No rules Recorded run tions)					
Drive	7 manoeuvres, lane 70/70 changes only when legal	.16	.157	231/231 ticks match the rules; 18/18 red lights, 3 lane changes, exit taken		
Doom (aiming)	retreat, shoot, turn left, 48/48 turn right, explore; no shoot at 0 ammo	.20	.729	arena 240/240, 8 kills; E1M1 200/200 (133 explore, 64 shoot, 3 turn), 4 kills		
Snake	safe moves	.984	.48	– 216/217 ticks, 24 food		

Drive. The rule list is stop, brake, change lane right, accelerate, change lane left, follow, and a hold speed default, with lane changes removed from the candidate set when the lane is occupied or leads away from the exit. The model matched the rules on 70 scripted situations, including 14/14 exit lane changes and 15/15 overtakes. Removing the guardrail and offering all seven actions drops agreement to .771.

Doom. The model picks one of five actions and the engine presses the key, with a closed-loop turn by the enemy’s bearing. On 48 scripted situations it matched the rules 48/48, including 29/29 turns, and it matched 20/20 sentences in the real engine’s vocabulary. In a recorded run on DOOM’s E1M1, with the engine walking a scripted route when the model chooses explore, all 200 decisions match the rule list: 133 explore, 64 shoot and 3 turns, for 4 kills.

Snake. The state is sentences such as “The snake must move right to close the larger gap to the food.” and “Moving left hits the snake’s body.”, the question is “Which move brings the snake closer to the food without dying?”, and the candidates are the safe moves. It matched the literal greedy policy on 61 of 62 boards (.984). Moving the same rules into the question as four direction rules without a default scored .82–.95 with $P(\emptyset)$.84–.94.

What the rendering controls. The same weights and the same rules succeed or fail depending on how we write the state, which makes these games a closed-loop instance of Section 6.3. Four regularities held across games.

- *Agent-subject sentences that pre-compute the comparison fire their rule.* “The player must turn left to face the enemy.” fires turn left 29/29; the object-location form “The nearest enemy is 5 cells to the left.” fired 0/12 over five phrasings. In Drive, “The car must take the exit on the right in 200 m.” fires 11/11, and “The exit on the right is 200 m ahead.” 0/11. In Snake, the pre-computed sentences score .984 against .774 for the coordinate form “The food is 3 cells to the right and 2 cells up.” on the same 62 boards.
- *An explicit default rule beats “none of the above”.* “explore: applies when the player sees no enemy” fires 6/6; “explore: applies when none of the above rules fire” fires 0/8, with the mass going to shoot and retreat.

- *Raw numbers the model would have to compare leak into the decision.* With “The player has 41 health and 34 ammo.” in the state, an imp in the crosshair reads `retreat .56 / shoot .42`; without the sentence `shoot` is .89. Mid-health crosshair situations score .500 with the numbers sentence and 1.000 without it, and 66 of 200 decisions in an earlier recorded run failed this way. Health and ammo now live on the game display, outside the state.
- *One condition per rule, within the trained list length.* A rule containing “or” fires where it should not (Drive .30 and .61 with a disjunctive rule vs. 1.00 without). A rule at position 6 fired 0/5 under four phrasings and the same rule at position 4 fired 4–5/5, matching the training shape of three rules plus a default.

A.13 Development chronology

Sections 3.3, 5 and 6 are organized by claim. The experiments ran in this order: (1) a candidate-blind decision-state branch (dual-encoder-style $Z(x, q)$ with an energy scorer over cached candidate embeddings), which produced the ladder of Table 6 and was closed by the shuffled-question control; (2) a candidate-in-suffix decision pass, comparing letter-logit, candidate-blind semantic, and contextual-candidate readouts, which produced Table 2; (3) the last-layer tap and rendered abstain-line diagnoses, which merged a planned two-regime (energy + native) architecture into a single model; (4) the workflow and uncertainty corpora, the mixture sweep, and null augmentation; (5) the Qwen3 size ladder at 1.7B/4B/8B/14B with frozen controls; (6) the 14B bundle (facts-first long-state rendering, `-drop_truncated`, DecisionMix v2, typed heads, likelihood-based checkpoint selection, frozen-teacher KD) and its matched no-KD control; (7) the Qwen3.5 port; (8) cluster bootstraps, the matched truncation ablation, and the purpose-built held-out long-state set, which produced the paired tests of Appendix A.11 and the 2×2 of Section 5.2; and (9) the v2 releases, `ts1c` (typical-small) and `tm2` (typical-medium), both trained on the facts-first corpus.

We closed the candidate-blind branch on Δ_q . Each student closed 20–55% of the chance-to-teacher accuracy gap, and the shuffled-question control showed that none of the closed gap was question-dependent; selection on accuracy would have kept the branch.

A.14 JevBench leaderboard context

On JevBench we drop $P(\emptyset)$ and renormalize the remaining probabilities; option order follows the harness (reversing it moves one checkpoint’s standard accuracy by ± 4 points); majority baselines are .311 standard, .284 easy and .336 hard. Table 20 places our numbers among other entries scored on the same 231 public ids. Entries differ in protocol and rendering, several have no method paper, and the 146 private judge items are not included. The public standard tier has $n_{\text{eff}}=36$ independent states, and rendering alone moves a frozen model by 12.5 points (Section 6.3), which exceeds most of the spread in this table (Section 4).

Table 20: Other entries scored on the same public 231 ids, from the harness’s own per-item files. JevBench (public subset) for all rows including ours. Majority/chance baselines: .311 standard, .284 easy, .336 hard. 95% cluster-bootstrap intervals for our own rows are in Tables 4 and 16; differences smaller than ± 6 –13 points (standard) or ± 9 points (hard) are not rankings.

Entry (backbone)	standard (72)	easy (48)	hard (111)
open-jev-deberta-v3-large (classifier)	.431	1.00	.378
GLiNER2	.639	1.00	.369
jeff (GLiFormer 400M)	.750	1.00	.387
Laya (ModernBERT)	.694	1.00	.351
open-alternative-jev (Qwen3.5-4B)	.833	1.00	.568
system-one-open (Gemma E2B LoRA)	.931	1.00	.486
SemIf (Qwen3.5-4B)	.986	1.00	.613
OpenJev (26B-A4B)	.972	1.00	.640
Jev 1.13.0 (closed)	.986	1.00	.730
Typical, Qwen3-14B (t11b)	.931	1.00	.450
Typical, Qwen3.5-4B (tm2, typical-medium v2)	.861	1.00	.495

A.15 Long-state truncation: distributions and the held-out evaluation set

Figure 4 gives the `data_wf_long` state-length histogram with truncation fractions at 256/1024/2048/3072 tokens, which is the measurement behind the 98.8% figure of Section 3.3, together with `long_policy` accuracy across the bug/fix sequence. The tokenizer’s `truncation_side` is ‘right’; we verified this empirically on the released tokenizer.

We built the held-out long-state evaluation set (`scripts/make_long_eval.py`, $n=605$) for the 2×2 of Table 3. It has zero exact-state overlap with either training corpus, and we dropped Case-stem overlaps. We score it at a 4,096-token window against states with $p50 = 1,965$ and $\max = 2,843$ tokens, so no item is truncated at evaluation in any of the eight cells. The two renderings differ only in whether the evidence block precedes or follows the request text; the items, labels, and candidate sets are identical. The set shares a generator family with the training data, and it measures the training defect of Section 5.2. Significance for the matched-configuration gap is a two-proportion test over the 605 paired items ($z = 6.2$, $p = 5 \times 10^{-10}$ at 1.7B; $p = 7 \times 10^{-12}$ at 14B). At 1.7B the gap is $+0.091$ [$+0.043, +0.139$] on `policy_permit` and $+0.138$ [$+0.086, +0.190$] on `action_select`.

Released checkpoints. We trained the v1 releases before the facts-first fix; the v2 releases replace them with post-fix checkpoints (Table 21). On facts-first long states `ts1c` gains 34.9 points over `ts1b` and `tm2` gains 33.8 over `tm1b`, while facts-last accuracy moves by -0.8 and $+1.3$. JevBench does not register the change: `ts1c` reads .708/.432 against `ts1b`’s .694/.432.

Table 21: Release checkpoints on the 605 held-out long states (4,096-token window, no truncation), identical items in both render columns. Majority-class floors: .334 overall, .612 on `policy_permit`. JevBench (public subset).

Release	Checkpoint	facts-first	facts-last	<code>policy_permit</code> (first)	JevBench std / hard
typical-small v1	<code>ts1b</code>	.598	.798	.536	.694 / .432
typical-small v2	<code>ts1c</code>	.947	.790	.948	.708 / .432
typical-medium v1	<code>tm1b</code>	.612	.866	.555	.806 / .423
typical-medium v2	<code>tm2</code>	.950	.879	.967	.861 / .495

A.16 Operations and reproducibility notes

Training and evaluation ran on rented RunPod H100 instances (SXM and NVL variants). For reproducibility, we record the infrastructure issues that affected timing. A pod created with an SSH-only startup flag never reached a ready state on the account used and billed regardless; we hit this several times before diagnosing it, and a fixed exposed port at boot avoided it. Stopping a pod without an attached persistent volume wipes it, so we wrote checkpoints and manifests to a network volume or pushed them to the Hugging Face Hub before stopping any pod. `tmux kill-session` does not terminate a timeout-wrapped child process, which once produced a stray duplicate training run sharing a GPU with the intended one; process matching over SSH must target the exact command, since a loose `pkill` pattern can match and kill the SSH shell itself before the target process. Two pods sharing one Python environment on the same network volume corrupted it under concurrent dependency resolution; per-pod environments with dependency syncing disabled avoided this. Every run’s manifest records the harness commit, checkpoint hash, and dependency lock file used. We compute every JevBench per-item comparison in this paper against the same fixed set of 231 public ids from one harness commit.

A.17 Full result tables

Tables 22 to 26 give the complete cross-checkpoint result tables for every checkpoint with a completed evaluation. `ts1b` and `tm1b` are the v1 releases; `tm2` is typical-medium v2, and `ts1c` (typical-small v2) is in Table 21.

A.18 Training corpora

Table 27 lists the seven training corpora of Section 3.3; Appendix A.10 gives their construction.

Table 22: Evidence and knowledge-classification accuracy by checkpoint.

Model	Checkpoint	CLINC-150	TREC-fine	HWU64	20NG	SNLI	MNLI	BoolQ	ANLI
Qwen3-1.7B	ts1b	.801	.508	.761	.540	.894	.859	.824	.497
Qwen3-4B	tm1b	.847	.414	.769	.588	.909	.861	.843	.544
Qwen3-8B	ladder_8b	.769	.486	.740	.452	.853	.831	.822	.468
Qwen3-14B	t11b	.827	.508	.760	.690	.911	.868	.882	.583
Qwen3.5-4B	tm2	.795	.480	.717	.579	.900	.855	.840	.554

Table 23: Knowledge probe (MMLU-Pro among- K , Δ_q^{sh}) and TruthfulQA MC1.

Model	Checkpoint	MMLU-Pro among- K	Δ_q^{sh}	TruthfulQA MC1
Qwen3-1.7B	ts1b	.343	.127	.257
Qwen3-4B	tm1b	.458	.193	.408
Qwen3-8B	ladder_8b	.302	.093	.277
Qwen3-14B	t11b	.498	.235	.481
Qwen3.5-4B	tm2	.429	.194	.414

DecisionMix v2 spreads its 60,605 training rows evenly over levels 1–6. A leak check against the public JevBench ids finds 0 of 231 hits.

A.19 Method and recipe detail

Suffix batching. A batch of suffixes against one cached state runs in chunks, and the suffixes do not attend to one another.

Readouts. Equation (1) is a bilinear form with a small learned correction on the elementwise product. N1, the standard multiple-choice interface read without generating, scores h_D against the output embeddings of the option letters. N2 embeds the candidate text with a frozen text-embedding model, which carries meaning and no slot identity. N2 and N3 share the scorer, the factored null and the feature normalization and differ only in the candidate vector. N1 and N3 differ in the readout and in the null form native to each; option shuffling during training is shared by all three.

Gate inputs. The Choice gate reads a vector z of permutation-invariant statistics over the valid candidates (maximum score, top-2 margin, mean score, log-mean-exp of the scores), the projected decision state, and the mean and variance of the projected candidate states v_j . It is symmetric in the pairs (s_j, v_j) . Equation (2) is implemented as logits $\ell_j = \log \text{softmax}(s/T)_j + \log(1 - r)$ and $\ell_\emptyset = \log r$, so one cross-entropy over $K + 1$ outcomes trains both parts, and property (i) holds because $(1 - r)$ multiplies every non-null candidate. A null rendered as a candidate line would compete for softmax mass that dilutes as K grows. The Noul gate has the form of Equation (2) with its own parameters; its score statistics come from $\log P(\text{yes})$ and $\log P(\text{no})$ and its state input is h_D , since no candidate span is rendered. Every Noul gate input is symmetric in the two labels.

Routing. A row takes the Bernoulli path if and only if its candidate set is exactly {yes, no}, Score rows take the ordinal target, and every other row is scored by the Choice path.

Cost. The candidate-blind scorer’s per-decision cost does not grow with the rendered candidate set.

Mixture sources. E is SNLI (Bowman et al., 2015), MNLI (Williams et al., 2018), ANLI (Nie et al., 2020a), BoolQ (Clark et al., 2019), and wide-vocabulary intent and topic sets including CLINC-150, Banking77 and HWU64 (Larson et al., 2019; Casanueva et al., 2020), spanning many label vocabularies at few rows per source so that the model must read the rendered candidates. K draws on nine sources, with TruthfulQA-MC1 (Lin et al., 2022) held out and a leakage audit against MMLU-Pro (Wang et al., 2024b; Hendrycks et al., 2020). W covers routing, eligibility, extraction, urgency, tool selection and long policy documents. The sampler weights and the sweep behind them are in Section 5.3.

Table 24: Held-out workflow and external floors. Rows marked *full* are the post-hoc `eval_wf` pass over every row at full state length (4,096-token window, `-limit 0`); rows marked *train-time* are the 1,500-row, 256-token pass emitted during training, which truncates the long external sets and is not comparable to the full pass. The distinction matters on the external sets: on `ts1b`, PagerDuty reads .560 at train time and .817 at full length, because the 256-token window removes the incident text. Numbers in parentheses after held-out score are NLL.

Model	Pass	Held-out noul	Held-out score (NLL)	Held-out style	PagerDuty (.792)	jevlogs (.697)	Mind2Web (.427)
<code>ts1b</code>	full	.715	.520 (1.01)	.859	.817	.710	.357
<code>tm1b</code>	full	.811	.528 (1.03)	.859	.838	.673	.544
<code>t11b</code>	full	.831	.623 (0.95)	.880	.878	.546	.563
<code>ladder_8b</code>	train-time	.681	.543 (1.72)	.882	.321	.501	.404
<code>tm2</code>	train-time	.858	.586 (0.97)	.842	.757	.518	.497

Table 25: JevBench (public subset) accuracy (Brier): 72 standard over 36 paraphrase-clustered states / 48 easy / 111 hard; see Tables 4 and 16 for intervals.

Model (checkpoint)	Standard (Brier)	Easy (Brier)	Hard (Brier)
Qwen3-1.7B (<code>ts1b</code>)	.694 (.40)	1.00 (.005)	.432 (.79)
Qwen3-4B (<code>tm1b</code>)	.806 (.30)	1.00 (.012)	.423 (.77)
Qwen3-8B (<code>ladder_8b</code>)	.667 (.55)	1.00 (.015)	.396 (.83)
Qwen3-14B (<code>t11b</code>)	.931 (.18)	1.00 (.005)	.450 (.66)
Qwen3.5-4B (<code>tm2</code>)	.861 (.22)	1.00	.495 (.72)
Qwen3.5-4B, frozen 3-shot	.764 (.36)	1.00	.495 (.60)
Qwen3.5-9B, frozen semif-render	.931 (.11)	1.00	.595 (.51)
Qwen3-14B, frozen 3-shot	.819 (.28)	1.00	.559 (.56)

Recipe corpora. DecisionMix v2 has 60,605 rows over 12 domains and six boolean conditions at compositional depths 1–6; depth-7 compositions, three whole domains, two rubric styles and one grammar per level are evaluation-only. `data_u` has 31,029 rows, all soft targets, uses the UNLI validation split, and each of its four generators holds out a parameter range. `data_wf_long` has 23,318 multi-paragraph policy documents (Appendix A.21); Section 5.2 isolates the evidence-first render with a render-crossed design. Null augmentation and the mixture act on the operating points measured in Sections 5.3 and 7.3, and Section 7.3 reports what the 14B selection bundle does.

A.20 Cached-state equivalence

An equivalence test on the real model checks that the cached-state pass of Section 3.1 computes the same decision as a forward pass on $\text{concat}(x, q, A)$. Cached-state decisions match the full-concatenation path to within 10^{-3} across suffix chunk sizes 1, 2, 3, 5 and 64, are consistent under permutation of the queries in a batch, and repeat exactly.

A.21 Long-state corpus statistics

States in `data_wf_long` have p10/p50/p90 lengths of 1,190, 1,845 and 2,490 tokens. In the original render, with the case facts after the policy body, right-truncation at a 1,024-token window removed the facts from 98.8% of rows, and at 256 tokens from all of them. The facts-first render places the case inside the first 8% of the text, and `-drop_truncated` removes any row longer than the training window.

Table 26: Single-decision research-harness latency (ms), $L_s=256$. The 1.7B, 4B and 14B rows are one pod and one PyTorch build; we timed the 8B row (marked *) on a separate pod.

Model (checkpoint)	$K=2$	$K=32$	$K=256$	Peak memory $K=2 \rightarrow 256$ (GB)
Qwen3-1.7B	45	46	106	6.4 $\rightarrow$ 9.3
Qwen3-4B	56	56	118	14.6 $\rightarrow$ 18.6
Qwen3-8B*	70	71	126	29.3 $\rightarrow$ 34.1
Qwen3-14B	60	62	157	51.6 $\rightarrow$ 57.5

Table 27: Training corpora. “Held out” counts whole families, domains, styles or parameter ranges excluded from training, distinct from a random validation split; Appendix A.1 maps these onto generalization axes. No corpus contains any of the 231 public JevBench ids.

Corpus	Family	Train rows	Held out	Role
data_v5	E	633,213	label wordings	Evidence (NLI, BoolQ) and wide-vocabulary classification
data_kb	K	147,447	TruthfulQA-MC1, GPQA	Closed-book multiple-choice knowledge
data_wf	W	72,006	3 of 15 families	Rubric-conditioned workflow (Noul, Choice, Score)
data_wf_hf	W	86,600	3 external test/demo splits	External workflow sets (forms, rules, distillation)
data_wf_long	W	23,318	–	Long-state (multi-paragraph policy) workflow rows
data_wh	W	60,605	3 domains, 2 styles, 1 grammar per level, level 7	DecisionMix v2 counterfactual rubric groups
data_u	U	31,029	4 synthetic parameter ranges	Soft-target uncertainty